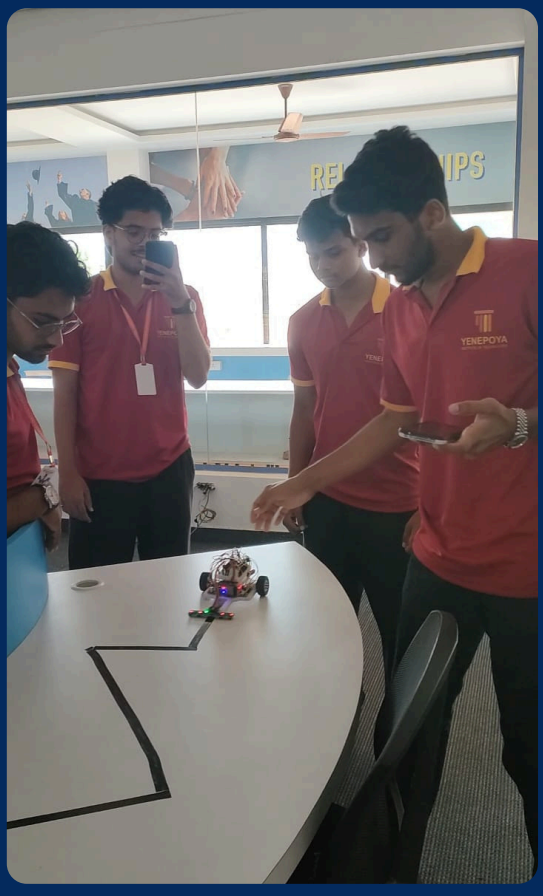
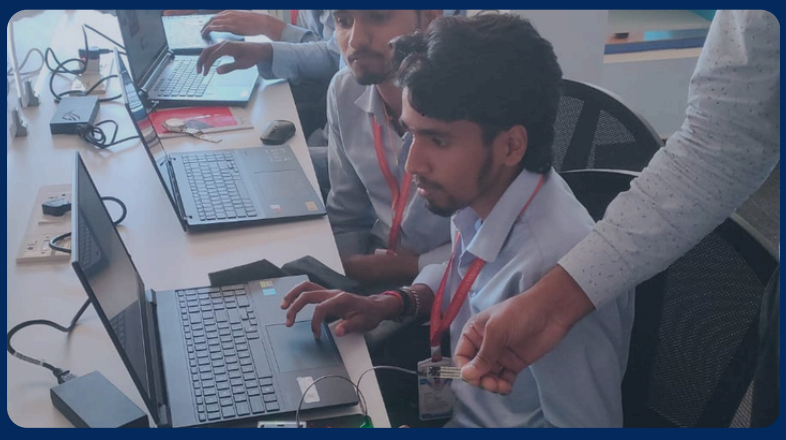
YENEPOYA
INSTITUTE OF TECHNOLOGY

inunity | innovation for community

INNOVATION AND DESIGNTHINKING COURSE

1st year students project booklet
2025-26





ABOUT

Yenepoya Institute of Technology (YIT)

Established in 2008, Yenepoya Institute of Technology (YIT) was founded with the visionary objective of fostering academic excellence and professional competence. Located in the vibrant educational hub of Moodbidri, Karnataka, the institute is dedicated to nurturing skilled professionals capable of thriving in the rapidly evolving global domains of Engineering and Technology.

Since its inception, YIT has achieved remarkable milestones across all facets of technical education. Underscoring its commitment to global benchmarks, the institution has been proudly awarded the prestigious Gold Rating by QS I- GAUGE for its excellence in higher education. The institute has earned a distinguished reputation among the premier engineering colleges in Karnataka, a testament to the high-quality education and practical training delivered by its deeply dedicated faculty, coupled with its state-of-the-art campus infrastructure. The programmes are accredited by NBA, New Delhi

This renowned institution is managed under the prestigious aegis of the Islamic Academy of Education-an organization globally recognized for its uncompromising commitment to quality education, ethical values, and academic leadership.

ABOUT THE

Innovation And Designthinking course

In this course students will learn how to design and prototype their ideas. Through a series of lectures and exercises students will learn and practise different prototyping techniques. This prototyping course emphasises on rapid prototyping, using prototyping device and system user interfaces, design considerations and perspective for devices. This course has topics that include design methods, modelling and simulation, design, and customization. This course on a whole, enables student transition from ideas to prototyping.

Program Objective

The program empowers students to transform ideas into functional prototypes that showcase innovation, technical skills, and real-world problem-solving.

KEY IMPACTS

120
HOURS OF TRAINING

288
STUDENTS PARTICIPATED

49
PROTOTYPES BUILT

TRL 3 - 49 PROJECTS

DEMO DAY :

PROTOTYPING BOOTCAMP

Prototyping Bootcamp is a hands-on session to learn, sketch, and rapidly build solutions using essential prototyping tools.

Project Content

Sl. No	Project Title	Page No
01	Smart food safety monitoring and hygiene management system for school and college kitchen	1
02	Problem in accident due to drink and drive and riders are not wearing helmet	2
03	Searching for parking in Public places result in time wastage and increases frustration and also leads to fuel wastages	3
04	Inefficient maintenance & underutilization of local manpower in Decentralized solar clusters	4
05	Smart Medication System Elderly Care	5
06	Sustaining Cleanliness on Beaches	6
07	Unhealthy Eating Habits in Children	7
08	Daily local bus Commuters lack clear & timely information about bus routes, stops and Schedules	8
09	Fire Outbreaks During Plane Crashes	9
10	3D Animation is too Expensive and Complex for Small Creators	10
11	Declining enrollment dropout risks and language based learning gaps threatening equitable education in Kannada medium schools of Karnataka	11
12	Poor Farming Decisions, Crop Losses and Reduced Income	12
13	DATA DRIVEN ALERT SYSTEM FOR FOR PEDESTRIANS AND TWO-WHEELERS ON HIGHWAY DUE TO POOR ROAD CONDITION	13
14	Software to Unreal Engine Integration Issues	14
15	Digital Hostel Management and Allocation System for Minority	15
16	Pepper Harvesting is Difficult Labor-Intensive and Risky for Farmers	16
17	Illegal Logging & Challenges in Sandalwood Agroforestry	17
18	Marine Pollution & Weakened Marine Ecosystem Resilience in Mangaluru	18

Project Content

Sl. No	Project Title	Page No
19	Enhancing Crowd Safety and Experience	19
20	Smart Cabin Climate Control System	20
21	Rethinking Accessibility in Beaches	21
22	Smart Offline Voice Based Farming Assistant for Farmers	22
23	Industrial Health and Safety Wearable	23
24	Communication Breakdown Between Deaf and Hard to Listen People	24
25	Controlling Wheel Chair Using BCI	25
26	Bio Medical Waste Management	26
27	Illegal Logging of Sandalwood Trees	27
28	Waste Segregation Issue	28
29	How Can We Make Public Mobile Charging Safe from Theft	29
30	Smart Food Safety Monitoring and Hygiene Management System	30
31	Smart Supply Chain for Auto Components in Mangalore	31
32	Smart Parking	32
33	Domestic Violence and Harassment of Women at Home Rising	33

Project Content

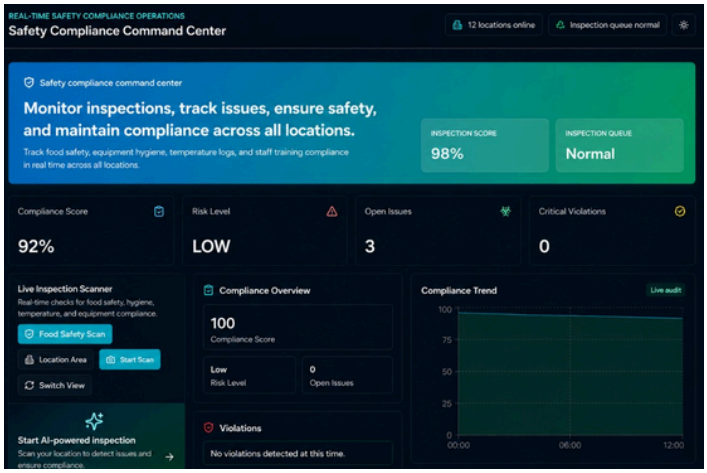
Sl. No	Project Title	Page No
34	Untreated Coloured Wastewater from Yarn-Dyeing Units	34
35	Root Cause Analysis of AI Model Selection Trust Issues	35
36	Crop Loss Due to Unseasonal Weather	36
37	Empowering Labour and Helping Society	37
38	Smart Ground Vibration Animal Repeller	38
39	Road Accidents Often Go Unnoticed for Critical Minutes, Delaying Emergency Response and Increasing Fatalities	39
40	Smart Green Tunnel	40
41	Unsafe Drinking Water	41
42	Nutrition Tracking	42
43	Smart Patient Monitoring and Alert System	43
44	Smart Nutrition and Anaemia Monitoring System for Children in Low Income Families	44
45	Leaf Classification for its Medicinal Use and Disease Detection	45
46	LIVE STOCK HEALTH MANAGEMENT SYSTEM	46
47	Smart AI Alarm Clock	47
48	Poor Crop Production and Water Wastage	48
49	Campus smart card and banking system	49

B.TECH

PROJECT - 01

Smart food safety monitoring & hygiene management system for school & college kitchen

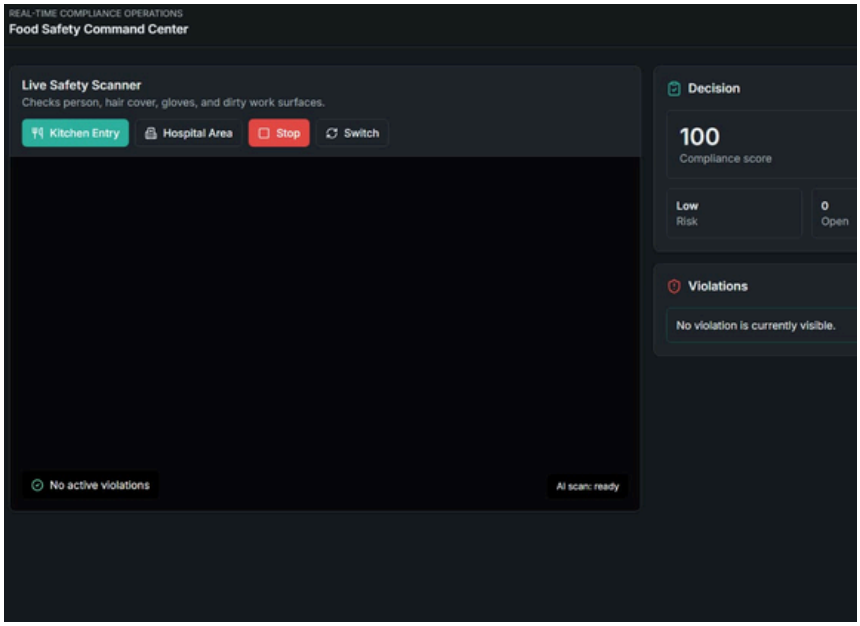
PROBLEM STATEMENT



Food safety and hygiene management in community and school kitchens are often inadequate due to manual monitoring, improper food storage practices, lack of hygiene compliance, and insufficient inspections. These issues can lead to food contamination and foodborne illnesses, posing serious health risks to students and community members. Therefore, there is a need for an affordable and scalable system that can monitor food storage conditions in real time and ensure adherence to hygiene practices.

IDEA GENERATION AND PROTOTYPE

The idea was developed to improve hygiene compliance in school and community kitchens through automated monitoring. A YOLO-based computer vision system was proposed to detect whether kitchen staff are wearing essential safety equipment such as gloves, hair caps, masks, and aprons. The prototype uses a camera and trained YOLO model to monitor staff in real time, identify hygiene violations, and generate alerts, providing an efficient and scalable solution for food safety management.



CONCLUSION

The proposed YOLO-based hygiene monitoring system provides an effective solution for improving food safety in school and community kitchens. By automatically detecting the use of gloves, hair caps, masks, and aprons, the system helps ensure hygiene compliance and reduces the risk of food contamination. The project demonstrates how computer vision can be used to create a practical, scalable, and efficient food safety management solution.

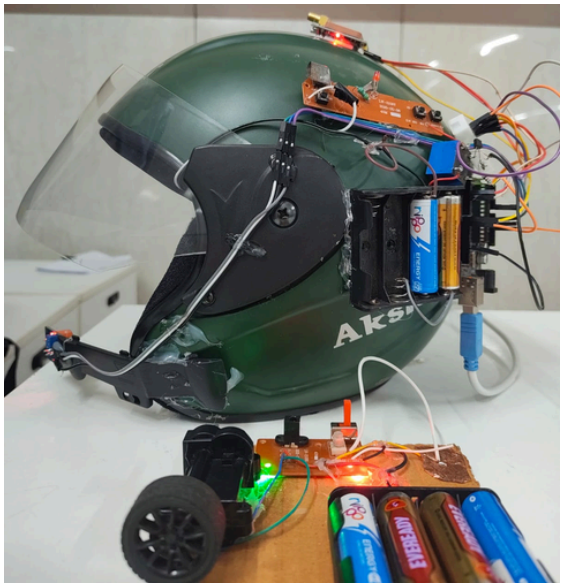
PROJECT - 02

PROBLEM IN ACCIDENT DUE TO DRINK AND DRIVE AND RIDERS ARE NOT WEARING HELMET

PROBLEM STATEMENT

Road accidents caused by drink-and-drive incidents remain a major safety concern, especially among two-wheeler riders. Existing safety measures are often unable to prevent intoxicated riders from operating vehicles. Therefore, there is a need for a smart helmet system that can detect alcohol consumption, prevent vehicle ignition when alcohol is detected, and provide real-time location tracking to enhance rider safety and reduce accident risks.

IDEA GENERATION AND PROTOTYPE



The idea for this project was generated by identifying the growing number of road accidents caused by alcohol-impaired riding. After analyzing various road safety solutions, a smart helmet system was proposed that could detect alcohol consumption and prevent vehicle operation. The prototype was developed using an Arduino Uno, MQ-3 alcohol sensor, GPS module, relay, and buzzer. The system detects alcohol in the rider's breath, disables vehicle ignition when alcohol is detected, and provides real-time location tracking. The components were integrated into a helmet prototype and tested successfully, demonstrating an effective solution for improving rider safety.

CONCLUSION

The Smart Helmet for Alcohol Detection and Safety successfully demonstrates an effective approach to reducing accidents caused by drink-and-drive behavior. By integrating alcohol detection, vehicle ignition control, warning alerts, and GPS tracking into a single system, the prototype enhances rider safety and promotes responsible driving practices. Testing results confirmed that the system accurately detected alcohol consumption, prevented vehicle ignition when necessary, generated warning alerts, and provided reliable location tracking. The project achieved its objectives and offers a practical, low-cost solution that can be further enhanced through IoT connectivity, emergency notifications, mobile applications, and cloud-based monitoring systems in the future.

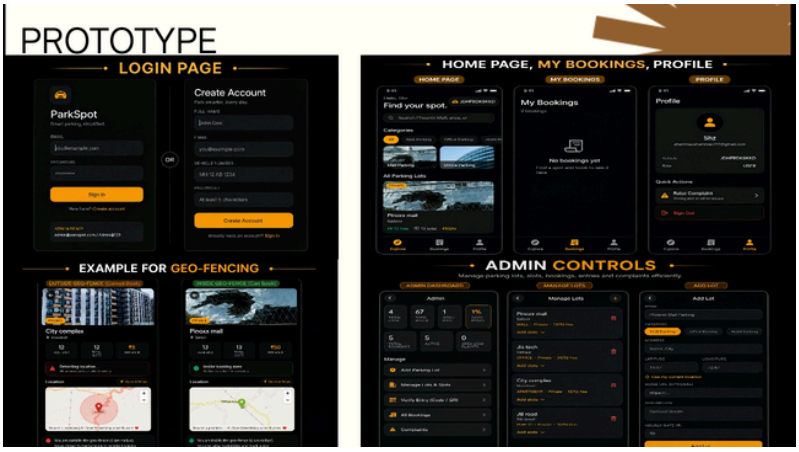
PROJECT - 03

SEARCHING FOR PARKING IN PUBLIC PLACES RESULT IN TIME WASTAGE AND INCREASES FRUSTRATION AND ALSO LEADS TO FUEL WASTAGES

PROBLEM STATEMENT

Drivers face difficulties in finding available parking spaces, resulting in traffic congestion, fuel wastage, and time loss. Existing parking systems do not provide real-time availability, booking options, or efficient parking management. This project aims to develop a Smart Parking Management and Booking System that simplifies parking search, reservation, and monitoring.

IDEA GENERATION AND PROTOTYPE



ParkSmart is a smart parking booking mobile application designed to simplify the process of finding and reserving parking spaces in urban areas. The solution addresses the growing challenge of parking congestion by enabling users to locate available parking slots in real time, make advance reservations, and verify bookings through QR code authentication. The application integrates GPS-based location services, real-time slot availability, secure booking management, and user-friendly navigation to enhance parking efficiency and reduce the time spent searching for parking spaces. The selection of ParkSmart is justified by its practicality, scalability, and ability to solve a common urban transportation problem while providing a seamless experience for both users and parking administrators.

CONCLUSION

The Smart Parking Management and Booking System was successfully designed and developed to address the challenges associated with finding and managing parking spaces in urban areas. The proposed system provides users with a convenient platform to search for parking locations, view real-time slot availability. reserve parking spaces, and verify bookings using QR codes. The integration of GPS-based location services, geo-fencing validation, and category-wise parking management improves parking efficiency and enhances the overall user experience.

PROJECT - 04

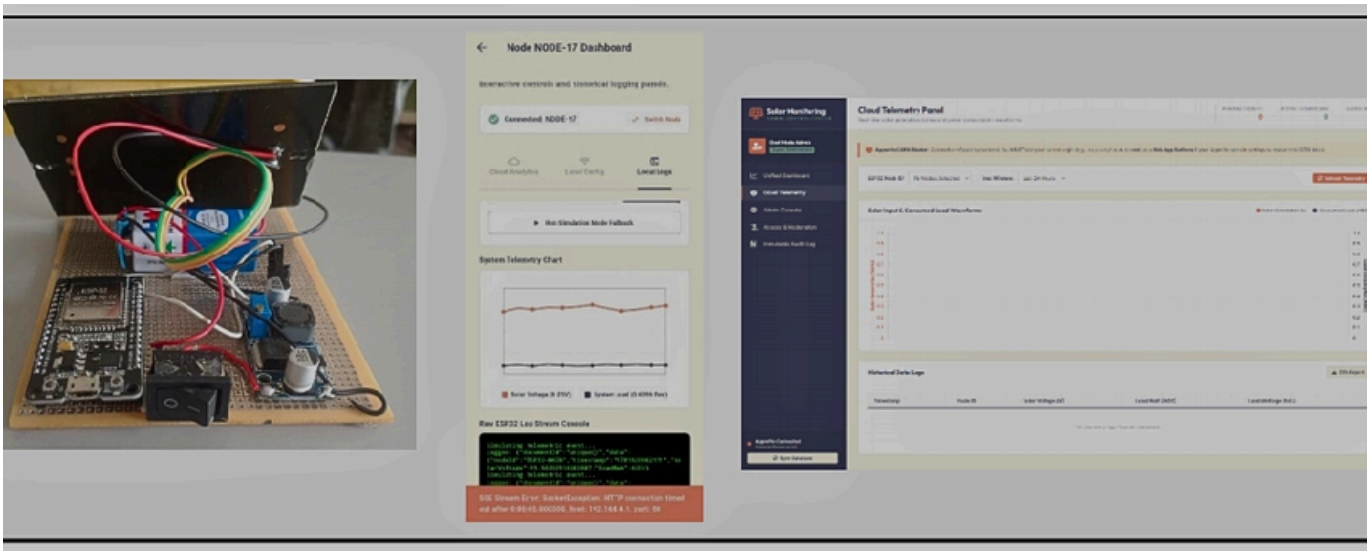
INEFFICIENT MAINTENANCE & UNDERUTILIZATION OF LOCAL MANPOWER IN DECENTRALIZED SOLAR CLUSTERS

PROBLEM STATEMENT

Rural off-grid solar maintenance in areas are highly inefficient due to the lack of a centralized digital support platform. When hardware fails, users with low digital literacy must rely on slow, informal communication networks, resulting in massive delays that hinder scalable, women-led local maintenance models.

IDEA GENERATION AND PROTOTYPE

To solve this, we engineered a cyber-physical system to automatically generate service tickets without user intervention. An ESP32 edge prototype actively monitors solar and battery metrics, compresses the data, and securely transmits it to a backend infrastructure that includes a mobile app for technicians and a centralized web administration console.



CONCLUSION

This project successfully transforms manual solar maintenance into an automated, datadriven ecosystem. By using an edge layer to handle diagnostics, it bypasses digital literacy barriers and provides technicians with verified service requests. Ultimately, this architecture empowers communities, supports women-led tech employment, and optimizes hardware longevity across decentralized grids.

SMART MEDICATION SYSTEM ELDERLY CARE

PROBLEM STATEMENT

Many elderly people forget to take medicines at the correct time due to memory issues and lack of supervision. Existing reminder systems only provide alerts and do not monitor medicine activity or inform caregivers. Hence, a smart IoT-based system is required to provide reminders, monitor medicine usage, and send alerts when doses are missed.

IDEA GENERATION AND PROTOTYPE

The idea is to develop an IoT-based smart medication reminder system using ESP32 as the main controller. RTC DS3231 is used for accurate timing, LCD displays medicine reminders, LED and buzzer provide alerts, and IR sensors detect medicine box activity. The Blynk IoT platform is used to send notifications to caregivers when medicine is not taken.

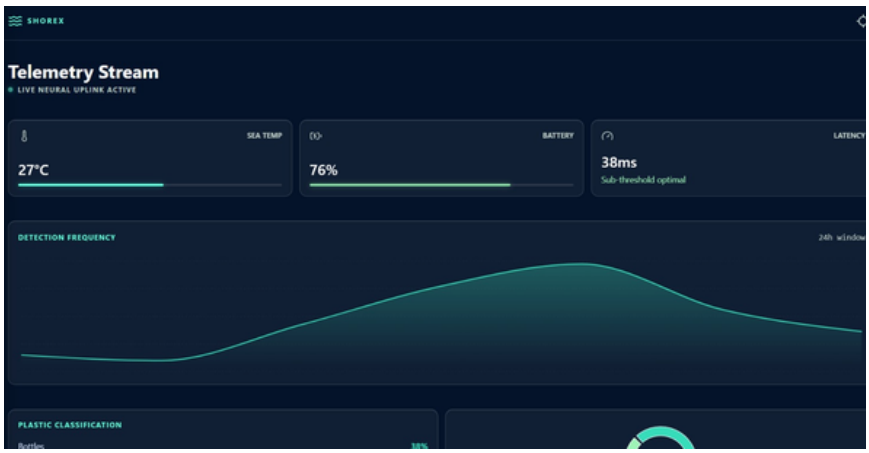
CONCLUSION

The developed system provides a low-cost and user-friendly solution for elderly medication management. It helps users take medicines on time and allows caregivers to receive alerts remotely. The project combines healthcare and IoT technology to improve safety and medication adherence.

PROJECT - 06

SUSTAINING CLEANLINESS ON BEACHES

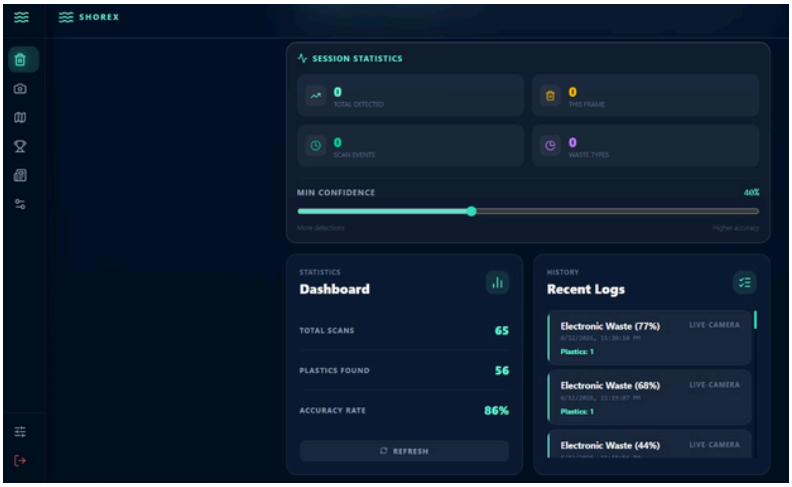
PROBLEM STATEMENT



Current plastic waste classification systems rely on expensive cloud infrastructure and internet connectivity, making them inaccessible and privacy-risky for community cleanup groups and rural schools. A fully on-device, serverless, browser-based AI system that enables real-time plastic detection with zero operational cost and complete offline functionality.

Limited to single-class plastic detection. Performance is bound by browser memory limits, mobile hardware variations, thermal throttling, and reduced accuracy for transparent plastics under reflective light.

IDEA GENERATION AND PROTOTYPE



Evaluated multiple concepts (including native apps and Raspberry Pi edge devices) using SCAMPER and mind mapping. A browser-based solution was chosen for its cross-platform compatibility, zero-installation requirement, and cost efficiency.

Designed an offline Convolutional Neural Network (CNN) based on MobileNetV2's depthwise separable convolutions to minimize size. Implemented as a Progressive Web Application (PWA), utilizing adaptive frame skipping to handle device thermal throttling.

CONCLUSION

The prototype successfully achieved real-time in-browser plastic detection with zero cloud costs. Testing on standard mobile devices delivered a viable inference latency of 50–200 ms and frame rates of 20–60 FPS. The current iteration cannot differentiate between plastic types (PET, HDPE, etc.), struggles with transparent plastics in poor lighting, and experiences performance drops on low-end hardware due to WebGL limitations or thermal throttling. Expanding the system to support multi-class plastic sorting, integrating an offline database for logging cleanup data, and exploring WebAssembly (WASM) to enhance processing speed on older devices.

UNHEALTHY EATING HABITS IN CHILDREN

PROBLEM STATEMENT

Childhood malnutrition and micronutrient deficiencies are common among school-going children in Karnataka, especially in rural areas. Parents and teachers often lack structured nutrition guidance, and planning balanced, affordable meals using locally available foods is difficult without clinical expertise. Existing solutions are not regional, not bilingual, and don't connect BMI assessment to actual meal planning.

IDEA GENERATION AND PROTOTYPE

We proposed NutriPrint, an AI-powered school nutrition platform built with Python FastAPI, Supabase (PostgreSQL), and the Groq LLM API. The system lets a teacher record a student's age, gender, height, and weight to instantly calculate BMI and classify it (Underweight/Normal/Overweight/Obese) using WHO/ICMR growth standards. Based on this result, an AI engine generates a personalised 7-day meal plan using a curated database of 53+ Karnataka regional foods, filtered by diet type, season, and budget (under ₹150/day). The platform also includes an AI Nutrition Advisor chatbot, a searchable food catalog, and a printable bilingual (English/Kannada) A4 poster with QR codes linking to step-by-step recipes — so the nutrition advice reaches parents at home, even without internet access. The prototype is fully deployed and live at

CONCLUSION

NutriPrint successfully demonstrates how AI and regional food data can bridge the gap between BMI screening and actionable nutrition guidance in schools. It is low-cost, scalable, bilingual, and requires no clinical training to operate, making it practical for real classroom adoption across Karnataka.

PROJECT - 08

DAILY LOCAL BUS COMMUTERS LACK CLEAR & TIMELY INFORMATION ABOUT BUS ROUTES, STOPS & SCHEDULES

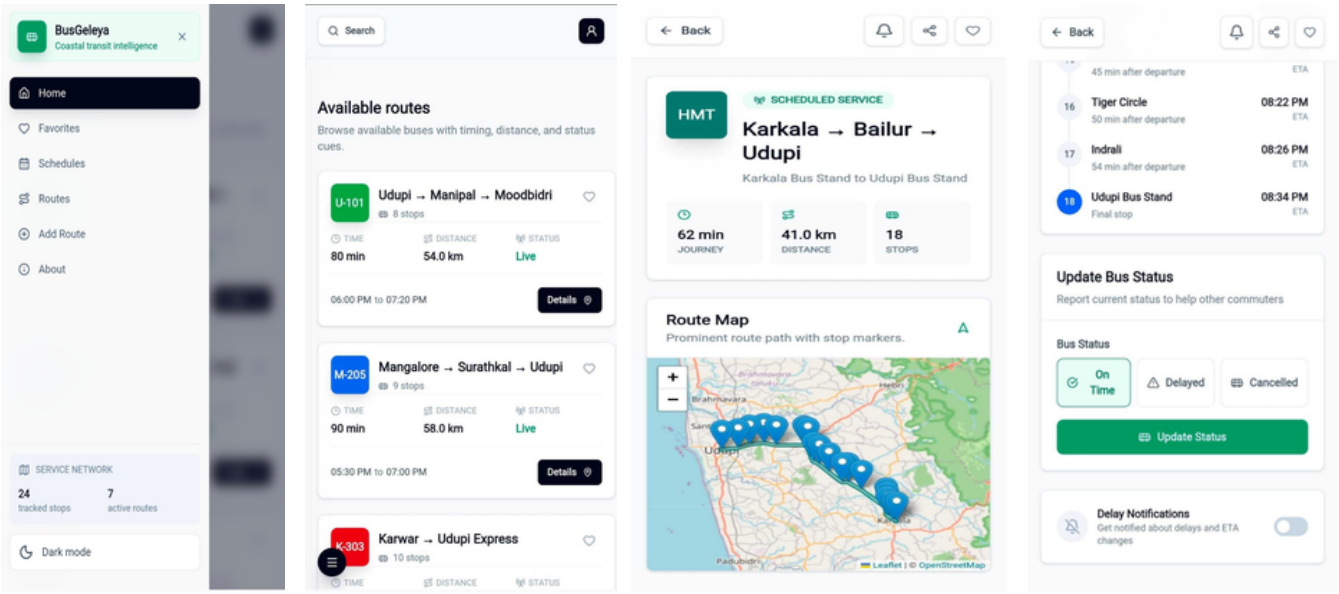
PROBLEM STATEMENT

Daily local bus commuters often face difficulties in obtaining accurate information about bus routes, stop locations, and arrival timings. This leads to confusion, longer waiting times, missed buses, and inefficient travel planning. Existing solutions are often fragmented or difficult to access, creating inconvenience for passengers who depend on public transportation.

IDEA GENERATION AND PROTOTYPE

The idea for BusGeleya was inspired by the common challenges faced by passengers while using local bus services. Through brainstorming sessions and the 5W1H analysis technique, the team identified the need for a simple and accessible bus information platform. Various solutions, including GPS-based tracking systems and IoT-enabled monitoring systems, were evaluated before selecting a route-based prediction approach due to its lower cost, scalability, and ease of implementation.

The prototype was developed as a web application using Next.js, TypeScript, Tailwind CSS, OpenStreetMap, and Leaflet.js. It allows users to search bus routes, view interactive maps with stops, check estimated arrival timings, and receive nearest-stop recommendations through a responsive and user-friendly interface.



CONCLUSION

BusGeleya successfully demonstrates a practical and cost-effective web-based bus information system that helps commuters access route details, stop information, and estimated arrival times without relying on continuous GPS tracking. The project improves accessibility to public transportation information, enhances commuter convenience, and provides a scalable foundation for future integration of real-time tracking and advanced prediction features.

SMART WATER SOURCE LOCATER FOR FIREFIGHTERS

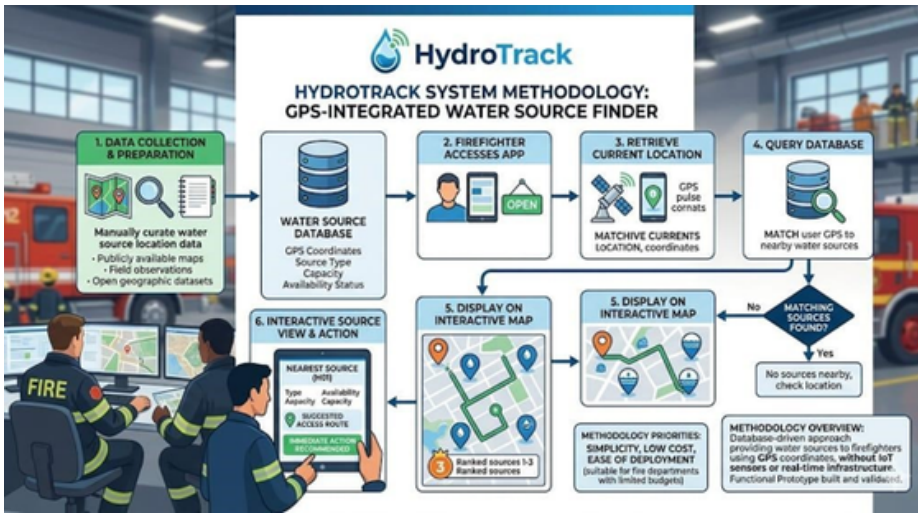
PROBLEM STATEMENT

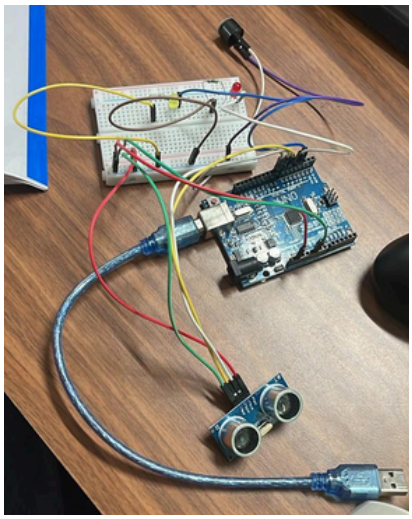
Firefighters responding to emergencies often face delays in locating nearby water sources such as hydrants, tanks, ponds, and reservoirs. Existing emergency response tools mainly focus on navigation and communication rather than water resource identification, which can reduce operational efficiency and increase risks to life and property. There is a need for a dedicated digital platform that provides quick access to water source locations, availability status, and proximity-based recommendations during firefighting operations.

IDEA GENERATION AND PROTOTYPE

The idea for HydroTrack was developed through brainstorming sessions and 5W1H analysis to identify challenges faced by firefighters in accessing water resources during emergencies. Multiple solutions such as IoT-based hydrant monitoring, crowdsourced reporting systems, offline map databases, and GPS-enabled platforms were evaluated. After considering feasibility, cost, deployment complexity, and impact, a GPS and GIS-based web platform was selected as the most practical solution.

The prototype was developed using Next.js, TypeScript, Tailwind CSS, OpenStreetMap, and Leaflet.js. The system enables firefighters to detect their GPS location, identify nearby water sources, view interactive maps with access routes, check source availability, and receive nearest-source recommendations through a responsive web interface.





CONCLUSION

HydroTrack successfully demonstrates a GPS-integrated, web-based water source locator that helps firefighters quickly identify and access nearby water resources during emergencies. The platform provides interactive mapping, proximity-based recommendations, and water source status information without requiring costly IoT infrastructure. The project validates the feasibility of a low-cost, scalable, and deployable solution that can be further enhanced through real-time IoT integration, municipal database connectivity, and advanced routing algorithms.

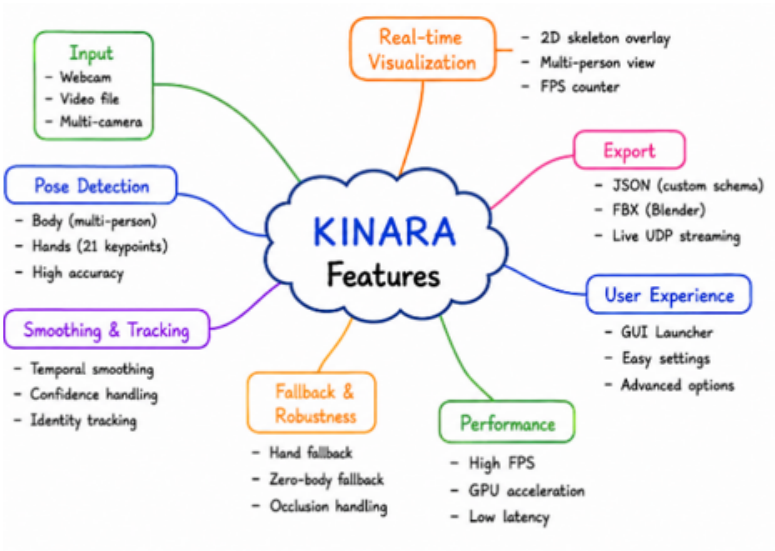
3D ANIMATION IS TOO EXPENSIVE AND COMPLEX FOR SMALL CREATORS

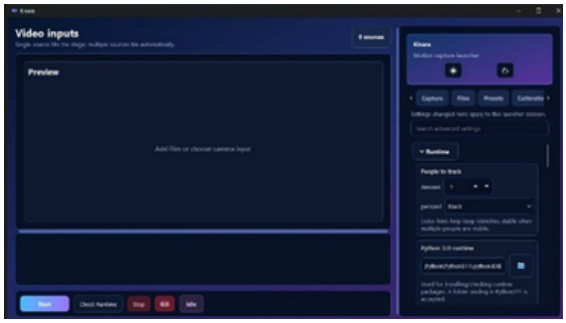
PROBLEM STATEMENT

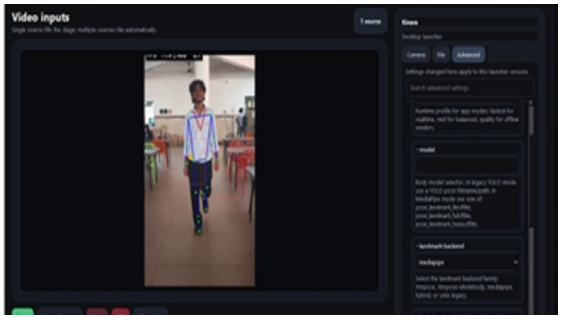
Creating realistic 3D animations usually requires expensive motion capture equipment and specialized hardware, making it inaccessible to students and independent developers. A low-cost and easy-to-use alternative is needed to convert human movements into 3D animations using standard cameras.

IDEA GENERATION AND PROTOTYPE

Various motion capture approaches, including depth sensors, wearable sensors, and RGB camera-based pose estimation, were evaluated. The prototype was developed using YOLO Pose, Python, OpenCV, and Blender. It captures human motion through a webcam, detects body landmarks, and maps them onto a 3D character to generate animations in real time.







CONCLUSION

The Human Motion to 3D Animation project successfully demonstrates a cost-effective motion capture solution using consumer-grade hardware and open-source technologies. The developed prototype accurately detects human body movements and converts them into corresponding 3D character animations without requiring specialized motion capture equipment. The project validates the feasibility of accessible motion capture systems for animation, gaming, and rehabilitation applications while providing a foundation for future enhancements such as improved occlusion handling, multi-camera support, and direct game-engine integration.

DECLINING ENROLLMENT DROPOUT RISKS AND LANGUAGE BASED LEARNING GAPS THREATENING EQUITABLE EDUCATION IN KANNADA MEDIUM SCHOOLS OF KARNATAKA

PROBLEM STATEMENT

Karnataka is witnessing a rapid and alarming decline in Kannada medium school enrollment. Over the last decade, student strength in government schools has fallen by nearly 19% — from 47.1 lakh in 2015–16 to 38.2 lakh in 2025–26. Between 2020 and 2026, approximately 3,617 government schools were shut down, with 2,670 closures in just the last academic year alone.

IDEA GENERATION AND PROTOTYPE

The idea was developed to bridge the language divide without abandoning cultural identity. After studying the root causes — poor digital infrastructure in Kannada schools, absence of mother-tongue tech education, and lack of career-focused language training — the team proposed Kannada Neva, a bilingual AI-powered learning platform.

The platform delivers STEM concepts in Kannada while simultaneously building functional English skills, eliminating the perceived trade-off between mother tongue and career readiness. Smart classroom integration and bilingual coding modules ensure students can learn robotics, science, and mathematics in their native language.

CONCLUSION

Kannada Neva successfully demonstrated that mother-tongue education and career-readiness are not mutually exclusive. By integrating AI-driven bilingual content delivery with modern curriculum standards, the platform offers a scalable, cost-effective model to reverse the decline of Kannada medium schools.

The project aligns with Karnataka's eight-point school reform roadmap announced in April 2026, and supports the broader vision of making government schools centers of excellence — preserving linguistic heritage while equipping students with the skills needed to thrive in a globalized economy.

POOR FARMING DECISIONS, CROP LOSSES AND REDUCED INCOME

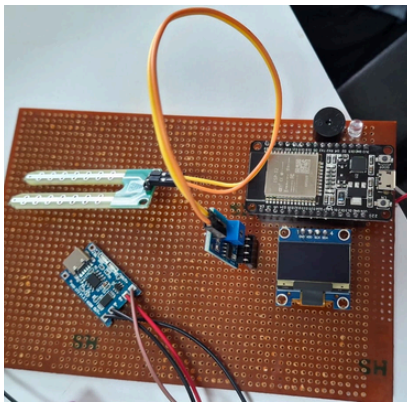
PROBLEM STATEMENT

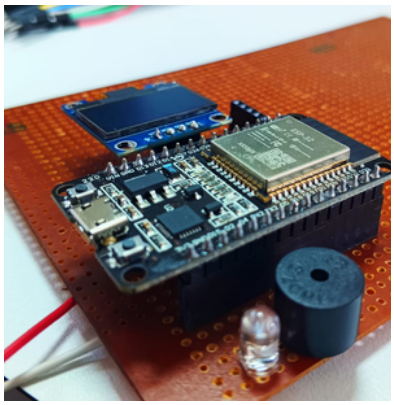
Small and marginal farmers in rural areas often receive weather hazard warnings too late or not at all, leading to significant crop losses. Existing agricultural applications require smartphones and internet connectivity, making them inaccessible to many farmers. Therefore, there is a need for a low-cost solution that can provide real-time weather alerts through SMS and voice calls using basic GSM mobile phones.

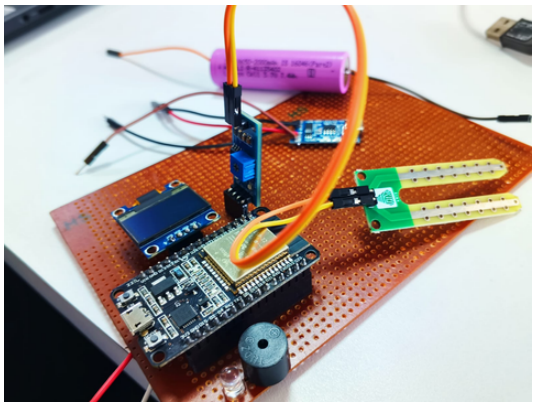
IDEA GENERATION AND PROTOTYPE

The idea was developed after identifying the lack of accessible weather warning systems for rural farmers. Through brainstorming sessions and 5W1H analysis, various communication technologies were evaluated, including LoRa, satellite messaging, and GSM-based systems. The final solution combined an ESP32 microcontroller, OpenWeatherMap API, SIM800L GSM module, Firebase, and a Flutter mobile application.

The prototype continuously monitors weather conditions, analyzes potential hazards using an AI-based decision engine, and automatically sends alerts through SMS, voice calls, Firebase notifications, and OLED display messages. A QR-based registration system was also implemented to simplify device setup and farmer onboarding.







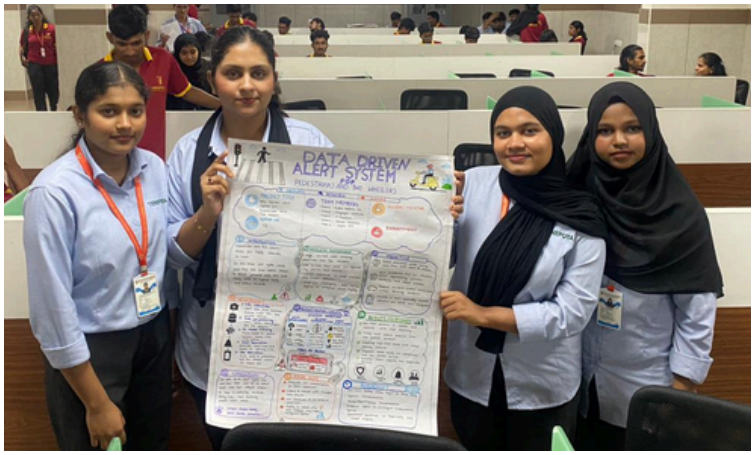
CONCLUSION

KRISHI AI successfully demonstrates a low-cost smart agricultural monitoring and alert system for rural farmers. By combining weather analysis, GSM-based communication, and cloud connectivity, the system provides timely weather warnings and supports better agricultural decision-making. The project offers an affordable and scalable solution for improving farm safety and reducing weather-related crop losses.

DATA DRIVEN ALERT SYSTEM FOR FOR PEDESTRIANS AND TWO-WHEELERS ON HIGHWAY DUE TO POOR ROAD CONDITION

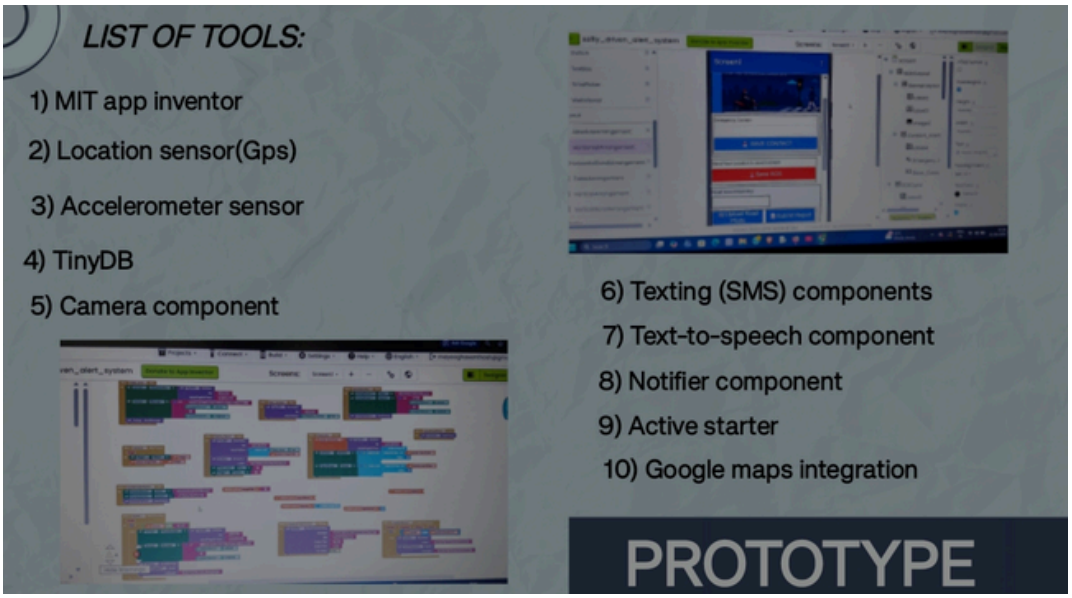
PROBLEM STATEMENT

How might we help two-wheelers, riders and pedestrians to travel safely from accident-prone and unsafe road conditions to a safer and more informed travel experience, so as to reduce accidents and improve road safety, despite poor infrastructure and lack of real-time alert systems?



IDEA GENERATION AND PROTOTYPE

The Data-Driven Alert System for Pedestrians and Two-Wheelers is a smart transportation safety platform designed to detect poor road conditions and provide real-time alerts to road users. The system collects road condition data through sensors, user reports, GPS mapping, and image-based analysis to identify hazards such as potholes, road cracks, waterlogging, and damaged surfaces. Key features include Hazard Detection, Real-Time Alert Notifications, Interactive Road Safety Map, Incident Reporting Module, Risk Level Classification, and Administrative Monitoring Dashboard.



CONCLUSION

The Data-Driven Alert System provides an effective solution for improving road safety for pedestrians and two-wheeler riders. By leveraging real-time data collection, AI-based hazard analysis, and instant alert mechanisms, the system helps users avoid dangerous road conditions and make safer travel decisions. The platform has the potential to reduce accidents, improve road awareness, and support transportation authorities in identifying and prioritizing road maintenance activities, ultimately contributing to safer and smarter highways.

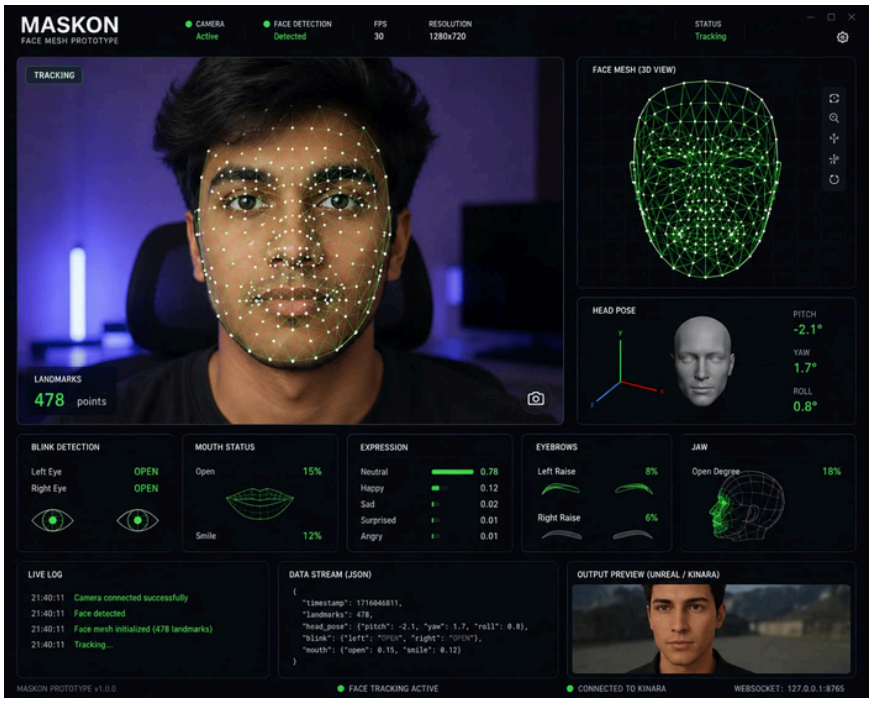
SOFTWARE TO UNREAL ENGINE INTEGRATION ISSUES

PROBLEM STATEMENT

Traditional motion capture systems require expensive cameras, specialized hardware, and dedicated studio environments, making them inaccessible to students, independent developers, and small production teams. There is a need for an affordable and scalable solution that can utilize multiple Android smartphones to capture, synchronize, process, and export motion data in real time.

IDEA GENERATION AND PROTOTYPE

The idea was developed to create a low-cost motion capture system using commonly available Android smartphones. Various motion capture approaches were studied, and a client-server architecture was selected. The prototype consists of multiple Android devices acting as motion capture nodes and a desktop application, Kinara.exe, serving as the central receiver and processing unit. The system captures motion data, synchronizes inputs from multiple devices, processes the data, and exports animation-ready outputs in formats such as FBX, BVH, and JSON.



CONCLUSION

The Kinara Multi-Device Capture System successfully demonstrates a smartphone-based motion capture solution capable of real-time data collection, synchronization, processing, and animation export. The project provides a cost-effective and scalable alternative to traditional motion capture systems, making motion capture technology more accessible for animation, gaming, and content creation.

PROJECT - 15

DIGITAL HOSTEL MANAGEMENT AND ALLOCATION SYSTEM FOR MINORITY

PROBLEM STATEMENT

Many educational institutions and minority welfare hostels still rely on manual methods for hostel admissions, room allocation, attendance tracking, and student record management. This often leads to delays, allocation errors, lack of transparency, paperwork burden, and difficulty in monitoring hostel occupancy. Students face challenges in applying for hostel accommodation and tracking application status, while administrators struggle to efficiently manage records and resources. Therefore, there is a need for a digital system that streamlines hostel management and allocation processes.

IDEA GENERATION AND PROTOTYPE

The Digital Hostel Management and Allocation System for Minority is a web and mobile-based platform designed to automate hostel administration and improve transparency in room allocation. The system enables students to submit hostel applications online, upload required documents, track application status, and receive allocation notifications. Administrators can manage student records, verify applications, allocate rooms based on predefined criteria, monitor hostel occupancy, and generate reports through a centralized dashboard.

The prototype consists of a Flutter-based mobile application and a web portal for administrators, supported by a Node.js backend and MongoDB database. The system includes secure login, online application submission, room allocation management, attendance tracking, complaint management, and notification services. Key modules include Student Registration, Hostel Application Management, Room Allocation Dashboard, Attendance Management, Complaint & Feedback System, and Admin Analytics Dashboard.

CONCLUSION

The Digital Hostel Management and Allocation System provides an efficient and transparent solution for managing hostel operations and student accommodation. By digitizing the application, allocation, and monitoring processes, the system reduces administrative workload, minimizes errors, and improves communication between students and hostel authorities. It enhances operational efficiency, ensures fair room allocation, and provides better record management, making hostel administration more organized, accessible, and user-friendly.

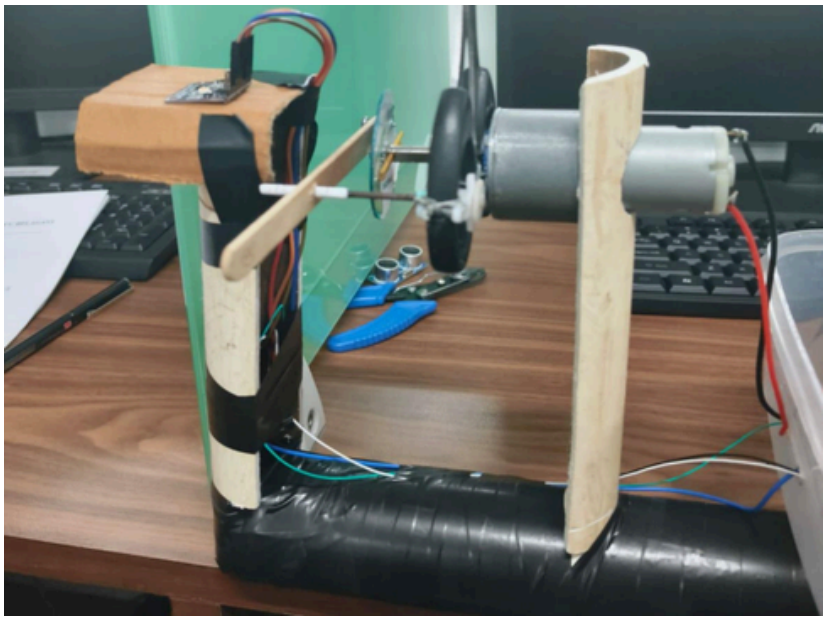
PEPPER HARVESTING IS DIFFICULT LABOR-INTENSIVE AND RISKY FOR FARMERS

PROBLEM STATEMENT

Pepper harvesting is traditionally performed manually, requiring significant labour, time, and physical effort. Labour shortages and increasing harvesting costs further reduce productivity, while existing automated harvesting systems are often too expensive for small-scale farmers. Therefore, there is a need for a cost-effective automated system that can assist farmers in detecting and harvesting ripe pepper efficiently.

IDEA GENERATION AND PROTOTYPE

The idea was developed to address labour-intensive pepper harvesting through affordable automation. Different approaches, including camera-based detection and sensor-based automation, were explored before selecting a combination of ESP32-CAM, TCS34725 colour sensor, ultrasonic sensor, relay module, and DC motor. The prototype detects ripe pepper based on colour, measures the distance to the target, and automatically activates a motorized harvesting mechanism when the required conditions are met. The system was integrated, programmed, and tested to validate its performance.



CONCLUSION

The Automatic Pepper Detection and Harvesting System was successfully developed and tested. The prototype effectively detected pepper and automatically activated the harvesting mechanism, reducing manual effort and improving harvesting efficiency. The project demonstrates the potential of low-cost automation technologies for smart agriculture applications and small-scale farming.

PROJECT - 17

ILLEGAL LOGGING & CHALLENGES IN SANDALWOOD AGROFORESTRY

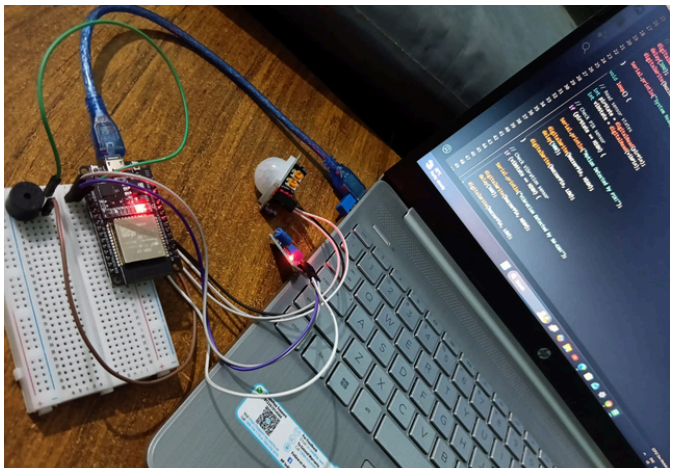
PROBLEM STATEMENT

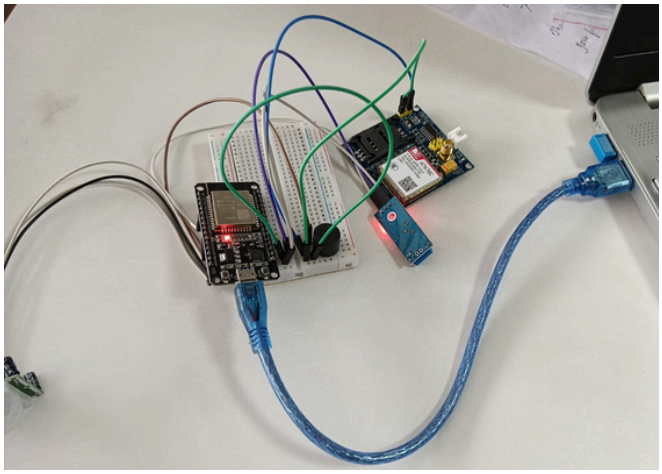
Illegal logging and unauthorized cutting of sandalwood trees have become a serious issue in agroforestry and forest conservation areas. Traditional monitoring methods rely on manual patrolling, which is time-consuming, labour-intensive, and unable to provide continuous surveillance. Delayed detection often results in environmental damage and economic loss, creating a need for an automated real-time monitoring system.

IDEA GENERATION AND PROTOTYPE

The idea was developed to improve forest security through automated monitoring. Different approaches such as manual surveillance, CCTV monitoring, GPS tracking, and IoT-based sensor networks were evaluated. After comparing cost, feasibility, and power consumption, an IoT-based sensor solution was selected.

The prototype was developed using an ESP32 microcontroller, PIR motion sensor, vibration sensor, GSM module, solar panel, and rechargeable battery. The system detects human movement and tree-cutting vibrations, processes the data using ESP32, and sends real-time SMS alerts to authorities through the GSM module. The solar-powered setup enables continuous operation in remote forest areas.





CONCLUSION

The project successfully developed an IoT-based monitoring system for detecting illegal sandalwood tree-cutting activities. The prototype accurately detected movement and vibration, generated real-time SMS alerts, and operated using a solar-powered energy system. The solution demonstrates a low-cost, energy-efficient, and reliable approach to protecting sandalwood resources and improving forest security.

PROJECT - 18

MARINE POLLUTION & WEAKENED MARINE ECOSYSTEM RESILIENCE IN MANGALURU

PROBLEM STATEMENT

Marine plastic pollution severely threatens the Mangaluru coast due to industrial discharge, fishing debris, and river runoff from Netravati and Gurupura rivers. Traditional manual surveys and cleanup drives are slow, inconsistent, reactive, and unable to differentiate high-risk waste types like ghost fishing nets from regular plastic. There is no automated, real-time system that can detect, classify, prioritize, and report marine waste effectively for use by KSPCB, NGOs, and cleanup teams.

IDEA GENERATION AND PROTOTYPE

After evaluating various approaches such as satellite imagery, drones, and basic mobile apps, we developed EcoVision — a Streamlit-based web application using a hybrid dual-model YOLOv8 architecture.

It combines surface and underwater detection models trained on merged datasets (TACO, Garbage6Class, and ICRA19 underwater frames). The system automatically detects the scene type, applies appropriate enhancements, runs parallel inference, and merges results with intelligent post-processing. Key features include context-aware Impact Scoring, Risk Assessment (marine life & human health), selective detection, severity heatmap, and one-click PDF report generation.

The prototype achieved mAP50 of 0.8075 on the V4 model.

CONCLUSION

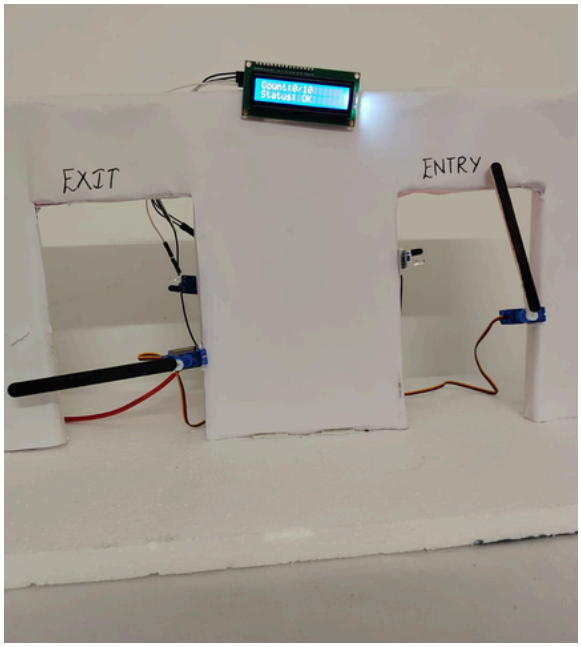
EcoVision demonstrates that an affordable, practical AI system for marine waste detection and reporting can be built using open-source tools and public datasets without expensive hardware.

The project's main contributions include a hybrid surface-underwater detection pipeline, context-aware risk intelligence, and an end-to-end reporting system tailored for Mangaluru's coastal needs.

While limitations exist in heavily turbid conditions, future scope includes GPS integration, mobile deployment, and expanded local datasets. EcoVision proves that AI can be effectively used for real environmental impact in coastal ecosystem protection.

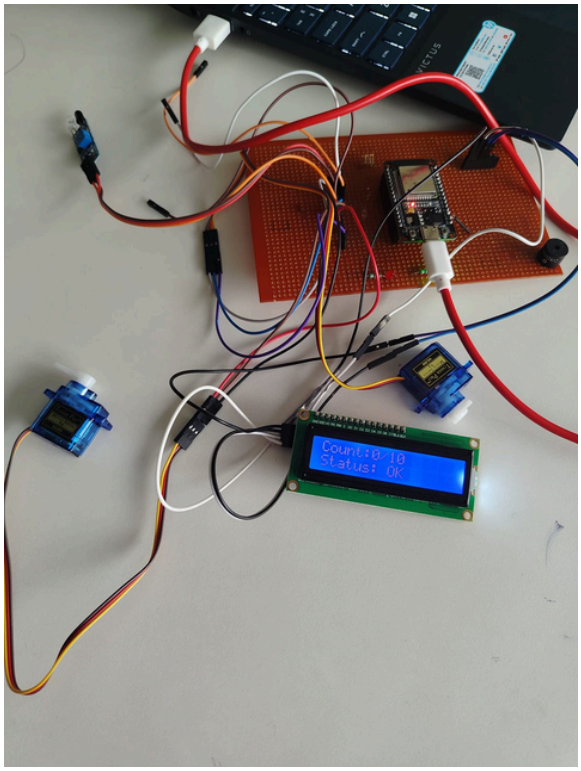
ENHANCING CROWD SAFETY AND EXPERIENCE

PROBLEM STATEMENT



Illegal logging and unauthorized cutting of sandalwood trees have become a serious issue in agroforestry and forest conservation areas. Traditional monitoring mePublic places such as shopping malls, hospitals, temples, railway stations, and event venues often experience overcrowding during peak hours. Manual crowd counting methods are inaccurate, time-consuming, and difficult to manage effectively. The lack of real-time occupancy monitoring and automated control increases the risk of congestion and safety issues, creating a need for an intelligent crowd monitoring and control system.thods rely on manual patrolling, which is time-consuming, labour-intensive, and unable to provide continuous surveillance. Delayed detection often results in environmental damage and economic loss, creating a need for an automated real-time monitoring system.

IDEA GENERATION AND PROTOTYPE



The idea was developed to address overcrowding in public places through automated monitoring and control. Various approaches, including manual counting, camera-based systems, and sensor-based techniques, were analyzed based on cost, accuracy, and implementation complexity. An IR sensor-based IoT solution using ESP32 was selected as the most practical approach.

The prototype was developed using ESP32, IR sensors, an LCD display, a servo motor, LEDs, a buzzer, and Firebase. The system automatically counts people entering and exiting a venue, displays the occupancy count in real time, updates a web dashboard, and controls gate access when the maximum capacity is reached. Alerts are generated through LEDs and a buzzer to prevent overcrowding.

CONCLUSION

The Smart Crowd Control and Monitoring System was successfully designed and tested. The system accurately monitored crowd occupancy, provided real-time updates through an LCD display and web dashboard, and automatically controlled entry when capacity limits were reached. The project demonstrates a practical, low-cost, and effective solution for crowd management and public safety.

PROJECT - 20

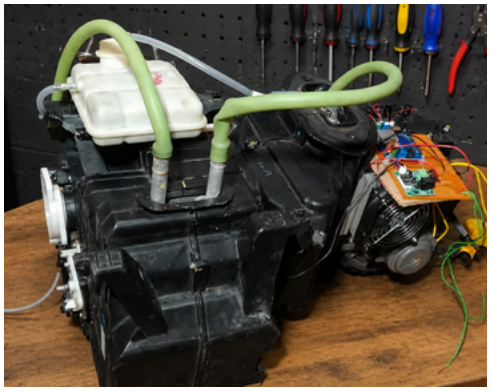
SMART CABIN CLIMATE CONTROL SYSTEM

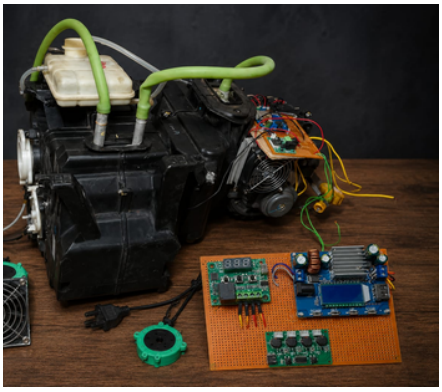
PROBLEM STATEMENT

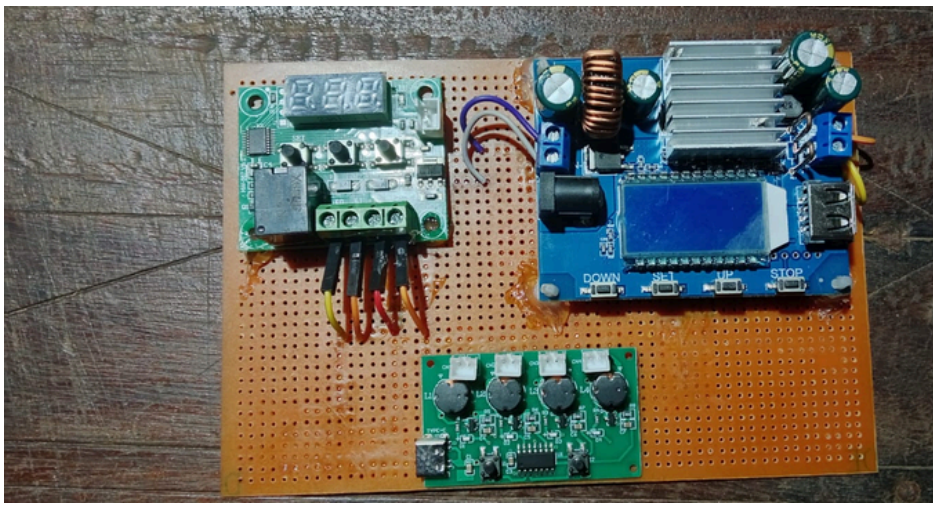
Conventional vehicle air conditioning systems consume a significant amount of fuel and increase carbon emissions. There is a need for an energy-efficient cooling system that can provide passenger comfort while reducing fuel consumption and environmental impact.

IDEA GENERATION AND PROTOTYPE

The idea was developed to create a compact cabin cooling system using thermoelectric (Peltier) modules. The system consists of a Peltier module, heat sinks, cooling fans, evaporator coil, coolant circulation pump, battery, and solar panel support. A prototype was designed and assembled to demonstrate cabin cooling without relying heavily on the vehicle's conventional AC system. The coolant absorbs heat from the cabin and circulates through the cooling unit, while the Peltier module provides the cooling effect.







CONCLUSION

The developed prototype successfully demonstrates an alternative cabin cooling method with lower energy consumption. The system has the potential to reduce fuel usage, improve energy efficiency, and support environmentally friendly transportation. Future improvements can enhance cooling performance and make the system suitable for real-world vehicle applications.

PROJECT - 21

ANXI-BOT:CAMPUS CALM CHATBOT

PROBLEM STATEMENT

College students often experience stress, anxiety, academic pressure, and emotional challenges that can affect their mental well-being and academic performance. Many students hesitate to seek help due to stigma, lack of awareness, limited access to counselors, or fear of judgment. As a result, emotional concerns may go unnoticed and escalate over time. Therefore, there is a need for an accessible and supportive platform that provides immediate guidance, emotional support, and mental wellness resources for students.

IDEA GENERATION AND PROTOTYPE

to support students by providing a safe and accessible platform for managing stress and anxiety. The chatbot enables students to engage in confidential conversations, receive emotional support, access stress-management techniques, and obtain recommendations for relaxation activities and campus wellness resources. Using Natural Language Processing (NLP) and sentiment analysis, the system identifies signs of stress, anxiety, or emotional distress and provides appropriate responses and coping strategies.

The prototype consists of An AI-powered chatbot engine processes user inputs and delivers personalized wellness guidance. The system also includes mood tracking, daily wellness check-ins, guided breathing exercises, motivational messages, and emergency support recommendations when high-risk emotional states are detected. Key modules include Chat Interface, Mood Tracker, Wellness Resources Page, Daily Check-In Dashboard, and Student Support Center.

CONCLUSION

Anxi-Bot: Campus Calm Chatbot offers a practical and accessible solution for promoting student mental well-being. By providing instant emotional support, mood monitoring, stress-management guidance, and wellness resources, the system helps students better manage anxiety and academic pressure. The platform encourages proactive mental health care, improves awareness of emotional well-being, and serves as a valuable support tool alongside professional counseling services, contributing to a healthier and more supportive campus environment.

PROJECT - 22

SMART OFFLINE VOICE BASED FARMING ASSISTANT FOR FARMERS

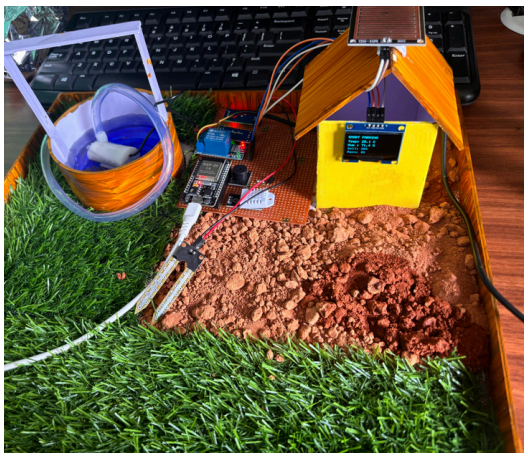
PROBLEM STATEMENT

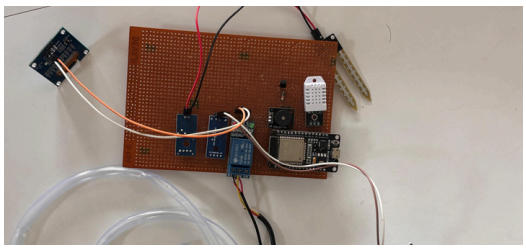
Farmers face challenges such as water scarcity, unpredictable weather conditions, rising cultivation costs, and limited access to timely agricultural information. Traditional irrigation methods often result in inefficient water usage and require continuous manual monitoring. Therefore, there is a need for an integrated smart agriculture system that combines real-time field monitoring, automated irrigation, and farmer assistance to improve productivity and resource management.

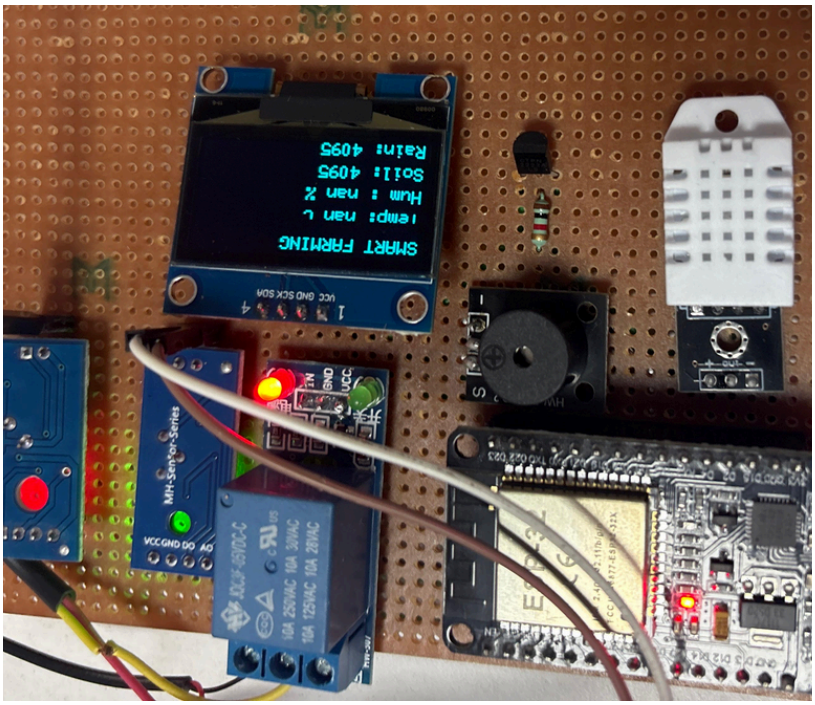
IDEA GENERATION AND PROTOTYPE

The idea was developed to address issues such as excessive water usage, manual irrigation, and lack of real-time monitoring. Through brainstorming and 5W1H analysis, an IoT-based solution was proposed using ESP32, soil moisture sensors, DHT22, rain sensors, a relay module, a water pump, and a mobile application with voice assistant support.

The prototype continuously monitors soil moisture, temperature, humidity, rainfall, and water levels. Based on soil moisture readings, the ESP32 automatically controls the water pump through a relay module. Sensor data is displayed on a mobile application, allowing farmers to remotely monitor field conditions and access agricultural information through a multilingual voice assistant.







CONCLUSION

The Smart Agriculture Monitoring and Irrigation System was successfully developed and tested. The system effectively automated irrigation, provided real-time monitoring of field conditions, and enabled remote access through a mobile application. The project demonstrated an efficient, reliable, and user-friendly solution for water management and smart farming practices.

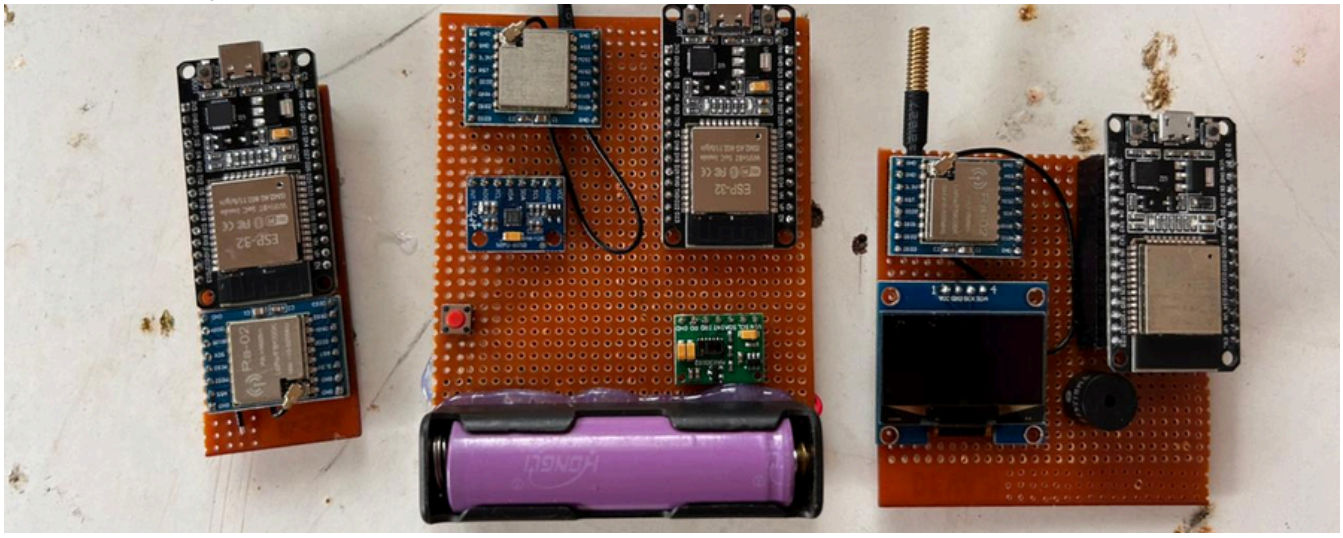
INDUSTRIAL HEALTH AND SAFETY WEARABLE

PROBLEM STATEMENT

Industrial workers, miners, rescue teams, and trekkers often operate in remote and hazardous environments where mobile network coverage is unavailable. During emergencies such as falls, injuries, gas leaks, fire accidents, oxygen deficiency, or health issues, communication delays can increase risks and hinder rescue operations. Existing solutions often rely on internet or cellular infrastructure, making them unsuitable for remote areas. Therefore, there is a need for a reliable, low-cost, and infrastructure-independent safety system for emergency communication and monitoring.

IDEA GENERATION AND PROTOTYPE

The project idea was developed by identifying communication and safety challenges faced by workers and rescue personnel in remote environments. Through brainstorming, mind mapping, and 5W1H analysis, a smart wearable safety system was proposed using ESP32 and LoRa Mesh Networking. The prototype integrates ESP32, LoRa SX1278 modules, MPU6050 sensors for fall detection, MAX30102 sensors for heart rate and SpO_2 monitoring, OLED displays, SOS emergency buttons, buzzers, and rechargeable batteries. The wearable device continuously monitors the user's condition and automatically generates emergency alerts during falls, abnormal health conditions, or SOS activation. Alerts are transmitted through relay nodes using a LoRa mesh network until they reach the base station, ensuring communication without internet or mobile connectivity.



CONCLUSION

The Industrial Health and Safety Control System was successfully designed, developed, and tested. The system accurately detected falls, generated emergency alerts, and transmitted messages through the LoRa mesh network without requiring internet or cellular connectivity. The prototype demonstrated reliable performance, low power consumption, and effective emergency communication, making it suitable for industrial safety, mining, trekking, and disaster management applications.

PROJECT - 24

INDUSTRIAL HEALTH AND SAFETY WEARABLE

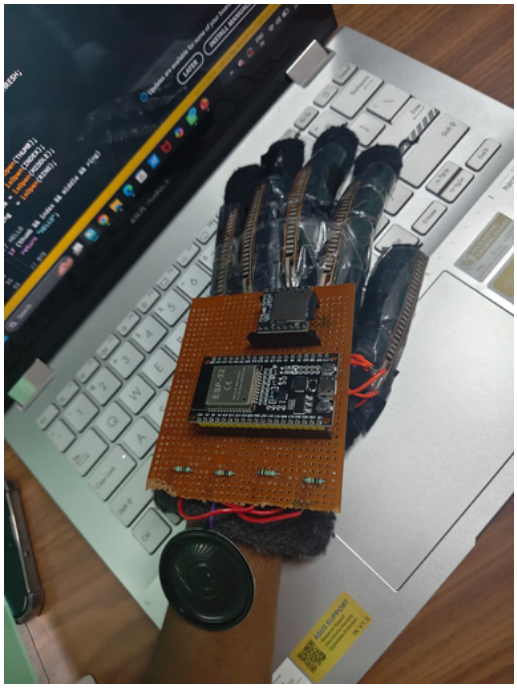
PROBLEM STATEMENT

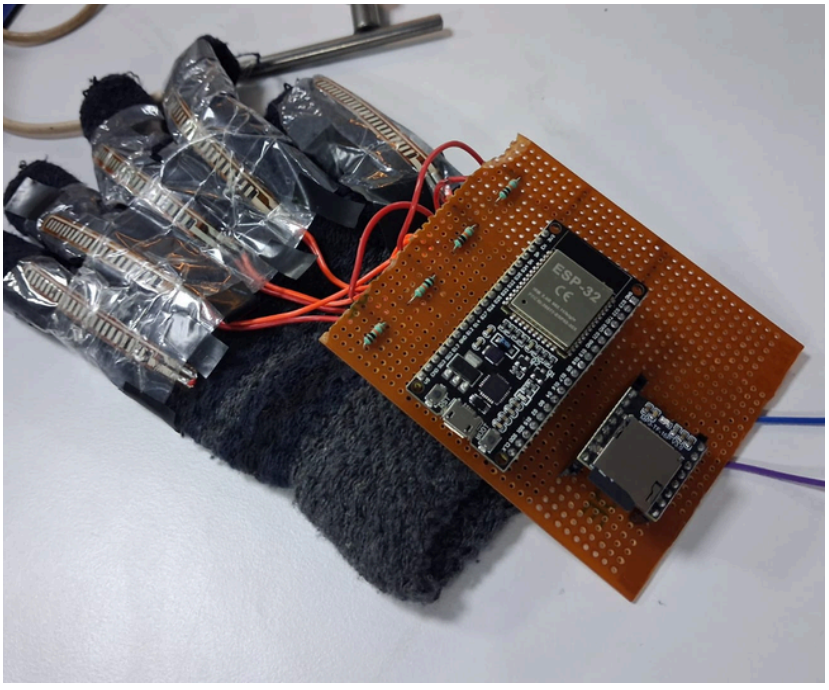
Individuals with speech and communication disabilities often face difficulties in expressing their thoughts, emotions, and needs. Most people do not understand sign language, making daily interactions challenging and leading to communication barriers, social isolation, and reduced accessibility. There is a need for an affordable, portable, and user-friendly communication system to assist speech-impaired individuals.

IDEA GENERATION AND PROTOTYPE

The idea was developed to help speech-impaired individuals communicate more effectively using hand gestures. After studying existing solutions such as sign language, text-to-speech devices, mobile communication apps, and wearable assistive devices, a smart glove-based communication system was proposed.

The prototype was developed using an ESP32 microcontroller, Flex Sensors, MPU6050, ADS1115 module, DFPlayer Mini, speaker, battery, and a wearable glove. The system detects finger movements and hand orientation, processes the gesture data through ESP32, and converts recognized gestures into predefined voice messages that are played through a speaker in real time.





CONCLUSION

The Communication Assistance Device successfully demonstrates an affordable and portable assistive technology for speech-impaired individuals. By converting hand gestures into spoken messages, the system improves communication, enhances accessibility, and promotes greater independence in daily interactions. The project also provides a foundation for future enhancements such as AI-based gesture recognition, multilingual support, and IoT connectivity.

PROJECT - 25

CONTROLLING WHEEL CHAIR USING BCI

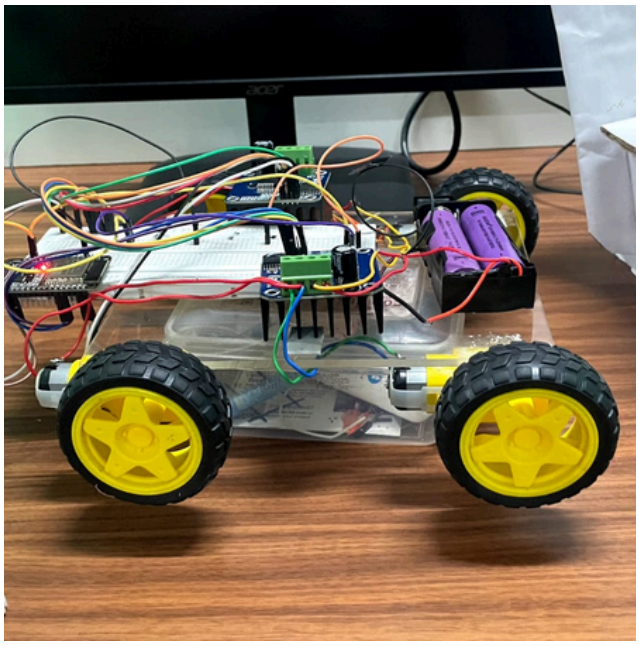
PROBLEM STATEMENT

Individuals suffering from paralysis, spinal cord injuries, ALS, muscular dystrophy, and other severe motor disabilities face significant challenges in achieving independent mobility. Existing wheelchairs often require joysticks, physical movement, or voice commands, which may not be suitable for many users. Therefore, there is a need for an affordable brain-controlled mobility system that can convert neural activity directly into wheelchair movement commands.

IDEA GENERATION AND PROTOTYPE

The idea was developed to provide independent mobility for individuals with severe physical disabilities using Brain-Computer Interface (BCI) technology. After studying existing mobility assistance systems, a brain-controlled wheelchair solution was proposed using EEG signal acquisition and embedded processing.

The prototype was developed using EEG electrodes, a BioAmp EXG Pill, ESP32 microcontroller, BTS7960 motor driver, DC motors, battery pack, and a wearable EEG headband. Brain signals are acquired through EEG electrodes, processed by the ESP32, and converted into movement commands such as Forward, Backward, Left, Right, and Stop. These commands control the wheelchair motors, enabling hands-free mobility through neural activity.



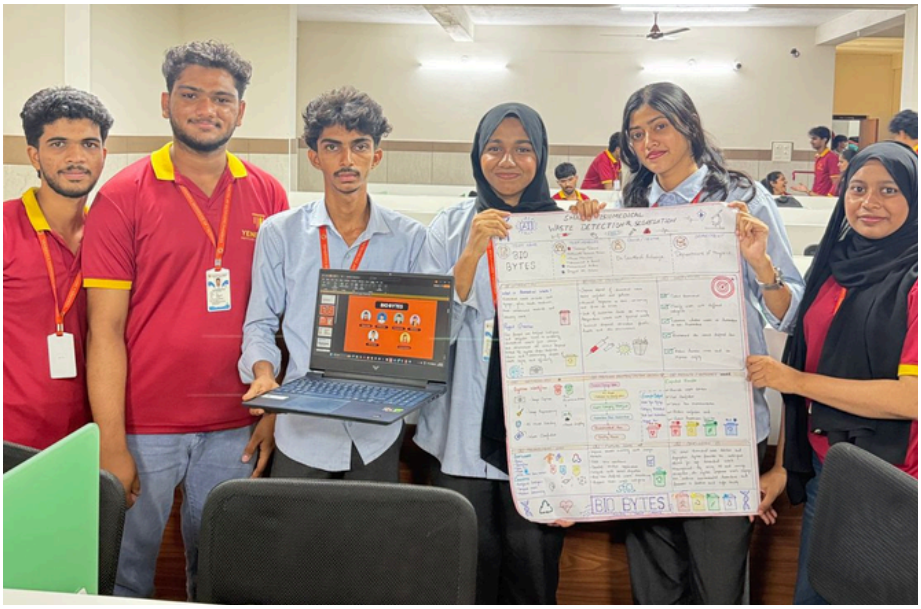
CONCLUSION

SynaptoLink successfully demonstrates the use of Brain-Computer Interface technology for assistive mobility. The system integrates EEG signal acquisition, embedded processing, and motor control to create a functional brain-controlled wheelchair prototype. The project highlights the potential of low-cost neuro-prosthetic solutions to improve independence and mobility for individuals with severe physical disabilities.

PROJECT - 26

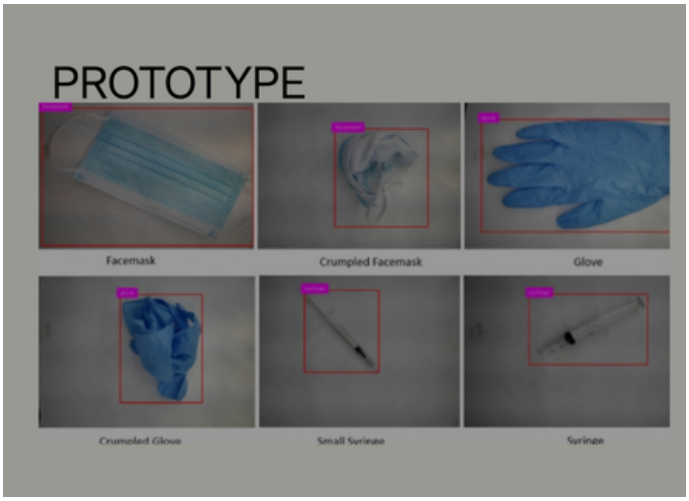
BIO MEDICAL WASTE MANAGEMENT

PROBLEM STATEMENT



Biomedical waste generated from hospitals, clinics, laboratories, pharmacies, and households is often disposed of improperly, especially in semi-urban areas. The lack of proper segregation, monitoring, and collection systems leads to environmental pollution, disease transmission, health hazards for waste handlers, and violations of waste management regulations. Existing manual systems are inefficient and do not provide real-time tracking or awareness regarding safe disposal practices.

IDEA GENERATION AND PROTOTYPE



During the brainstorming phase, the team explored several approaches to improve biomedical waste management:

-Smart waste bins with sensors
-Mobile application for waste collection requests
-GPS-based waste tracking system
-AI-powered waste detection and classification
-Real-time monitoring dashboard for authorities

After evaluating feasibility, cost, and effectiveness, the team selected an AI-based Biomedical Waste Management System that combines waste detection, classification, tracking, notification, and monitoring into a single platform.

CONCLUSION

The Safe Biomedical Waste Management System provides an intelligent and efficient solution for biomedical waste handling. By integrating AI-based waste detection, real-time tracking, automated notifications, and monitoring tools, the system improves waste segregation accuracy, reduces environmental pollution, minimizes health risks, and supports regulatory compliance. The project demonstrates how modern technologies can enhance public health and create safer and cleaner communities through effective biomedical waste management.

ILLEGAL LOGGING OF SANDALWOOD TREES

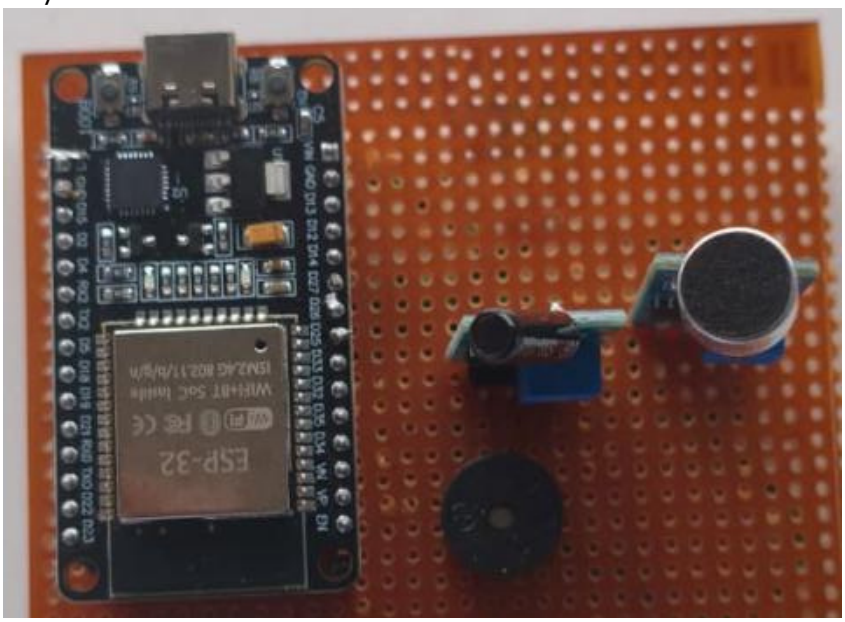
PROBLEM STATEMENT

Illegal cutting and theft of sandalwood trees result in significant economic losses and environmental damage. Existing monitoring systems are expensive, require continuous supervision, and often fail to provide immediate alerts. Therefore, a smart and automated protection system is needed to detect tree-cutting activities and notify users in real time.

IDEA GENERATION AND PROTOTYPE

The idea was generated through brainstorming sessions to address the growing problem of sandalwood theft in plantations and forest areas. Various solutions such as CCTV surveillance, GPS tracking, motion sensors, GSM alerts, and IoT-based monitoring systems were evaluated. The final solution selected was an IoT-based Smart Sandalwood Tree Protection System using sound and vibration sensors.

The prototype was developed using an ESP32 microcontroller, KY-038 sound sensor, SW-18010P vibration sensor, buzzer, rechargeable battery, and Wi-Fi communication. The sensors continuously monitor cutting-related sounds and vibrations. When abnormal activity is detected, the ESP32 processes the sensor data, activates a local buzzer alarm, and sends a real-time Wi-Fi alert notification to the owner or forest authority.



CONCLUSION

The Smart Sandalwood Tree Protection System successfully achieved its objective of providing real-time protection against illegal tree cutting. By integrating sound sensors, vibration sensors, ESP32, and Wi-Fi communication, the system offers a low-cost and effective solution for safeguarding valuable sandalwood trees. The project demonstrates the practical application of IoT technology in environmental conservation and sustainable forest management.

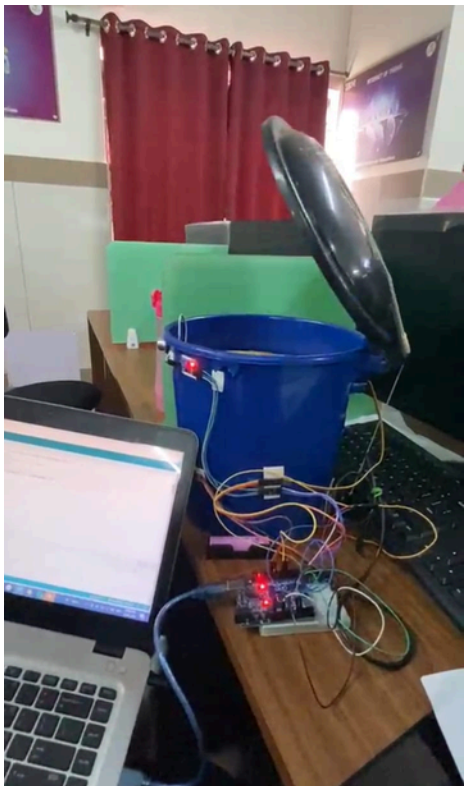
WASTE SEGREGATION ISSUE

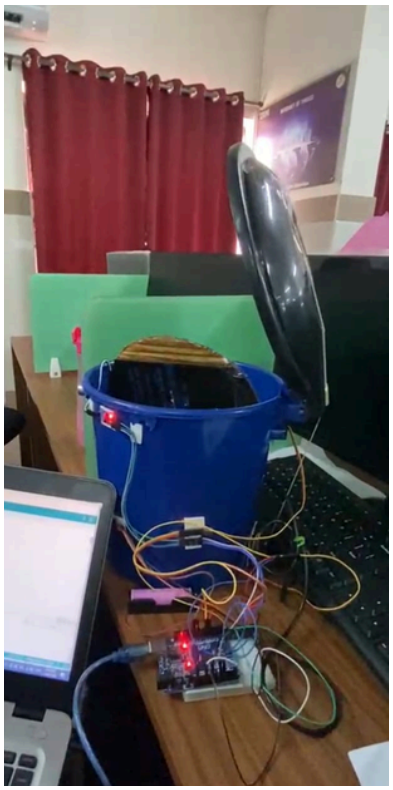
PROBLEM STATEMENT

Improper waste segregation at the source leads to reduced recycling efficiency, environmental pollution, and health hazards. Manual sorting of waste is time-consuming, unhygienic, and prone to errors, while existing automated systems are often expensive and complex. Therefore, there is a need for a simple, affordable, and automated solution that can efficiently classify and segregate waste at the point of disposal.

IDEA GENERATION AND PROTOTYPE

The idea originated from the growing need for efficient and low-cost waste management solutions. After reviewing existing smart waste management systems, a sensor-based automated waste segregation system was proposed. The prototype was developed using an Arduino Uno, HC-SR04 ultrasonic sensor, inductive sensor, soil moisture sensor, and MG995 servo motor. The ultrasonic sensor detects incoming waste, while the inductive and moisture sensors classify the waste as metallic, wet, or dry. Based on sensor inputs, the servo motor automatically directs the waste into the appropriate compartment.





CONCLUSION

The Smart Waste Segregation System successfully automated waste detection and classification using sensor technology and Arduino. The prototype demonstrated reliable performance in identifying metallic and wet waste while reducing manual effort and improving hygiene. The project proves that low-cost embedded systems can provide an effective solution for smart waste management and can be further enhanced with IoT and AI-based technologies.

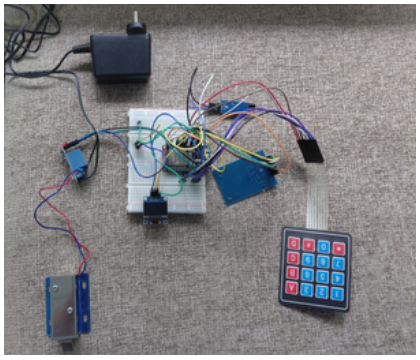
PROJECT - 29

HOW CAN WE MAKE PUBLIC MOBILE CHARGING SAFE FROM THEFT

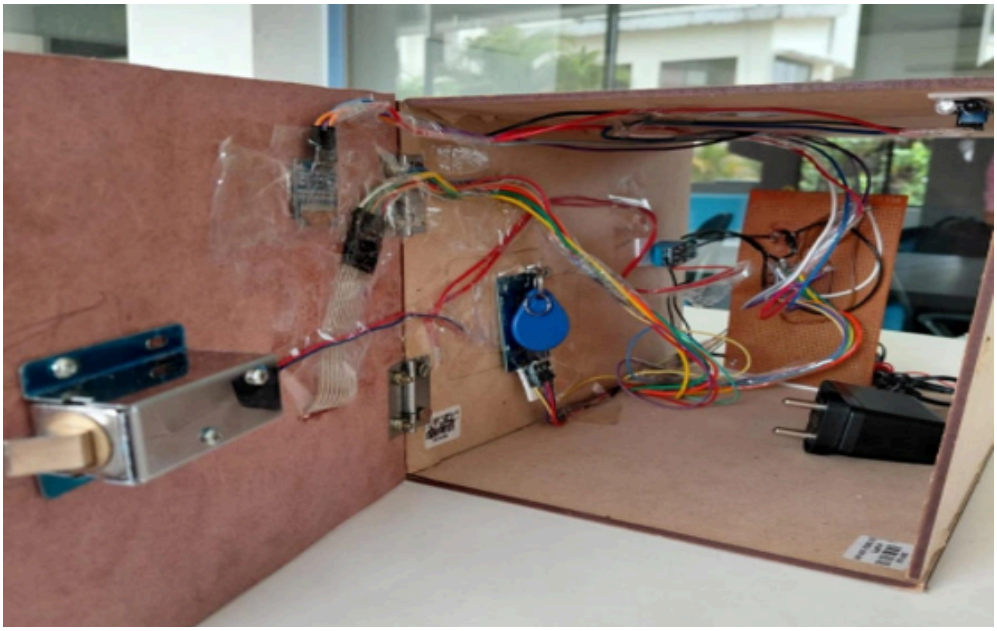
PROBLEM STATEMENT

With the increasing use of smartphones and portable electronic devices, users often face difficulties finding secure charging facilities in public places. Traditional charging stations allow devices to be charged but do not provide adequate protection against theft, unauthorized access, or damage. Therefore, a secure and efficient Smart Charging Locker system is required to enable users to safely store and charge their devices simultaneously.

IDEA GENERATION AND PROTOTYPE



The idea was developed to address the need for secure mobile charging in colleges, malls, airports, and other public spaces. The prototype was built using an ESP32 microcontroller, RFID reader, keypad, OLED display, relay module, IR sensor, and solenoid lock. Users authenticate themselves through RFID or a PIN, after which the locker unlocks, allowing the device to be stored and charged securely. The system ensures controlled access, real-time status display, and reliable charging.



CONCLUSION

The Smart Charging Locker was successfully designed and implemented, providing secure storage and efficient charging for electronic devices. RFID-based authentication, electronic locking, and charging modules worked reliably, ensuring device safety and user convenience. The system offers a practical, cost-effective solution for public and institutional environments and can be further enhanced with IoT, biometric authentication, and mobile app integration.

PROJECT - 30

SMART FOOD SAFETY MONITORING AND HYGIENE MANAGEMENT SYSTEM

PROBLEM STATEMENT

"Food facilities lack real-time automated monitoring to detect hygiene violations among staff — leading to preventable food contamination outbreaks."

Current Gap: Manual inspection is inconsistent, unreliable, and cannot run 24/7, leaving kitchens highly vulnerable.

600 MILLION: Foodborne illnesses globally each year (WHO 2022).

40% OF OUTBREAKS: Usually linked back to poor staff personal hygiene.

ZERO SOLUTIONS: Affordable, automated monitor essentials for small kitchens

IDEA GENERATION AND PROTOTYPE

To address this challenge, we developed HygieneGuard, a real-time AI-powered food safety compliance monitoring system. The solution utilizes computer vision and edge computing technologies to automatically detect multiple hygiene violations without requiring expensive GPU infrastructure. The system can identify critical violations such as the absence of face masks (93% accuracy), face touching (88% accuracy), hair touching (85% accuracy), and the absence of gloves (82% accuracy). Built using Python, Flask, MediaPipe, OpenCV, SQLite3, and Chart.js, the system captures live webcam feeds, processes video frames through an AI-based hygiene verification engine, stores violation records and snapshots in a database.

CONCLUSION

In conclusion, HygieneGuard successfully demonstrates an affordable and effective approach to automated food hygiene compliance monitoring. The system achieved real-time multi-class violation detection, processed lightweight webcam feeds on standard CPU-based devices, and validated over 1,771 recorded hygiene violations within its SQLite-based logging framework. The integrated dashboard provides comprehensive analytics, log management, and database control features, while real-time warning mechanisms enhance workplace safety. In the future, the system can be expanded with advanced AI models, cloud-based monitoring, multi-camera support, predictive analytics, and integration with food safety regulatory frameworks to further improve hygiene compliance and reduce foodborne disease outbreaks.

PROJECT - 31

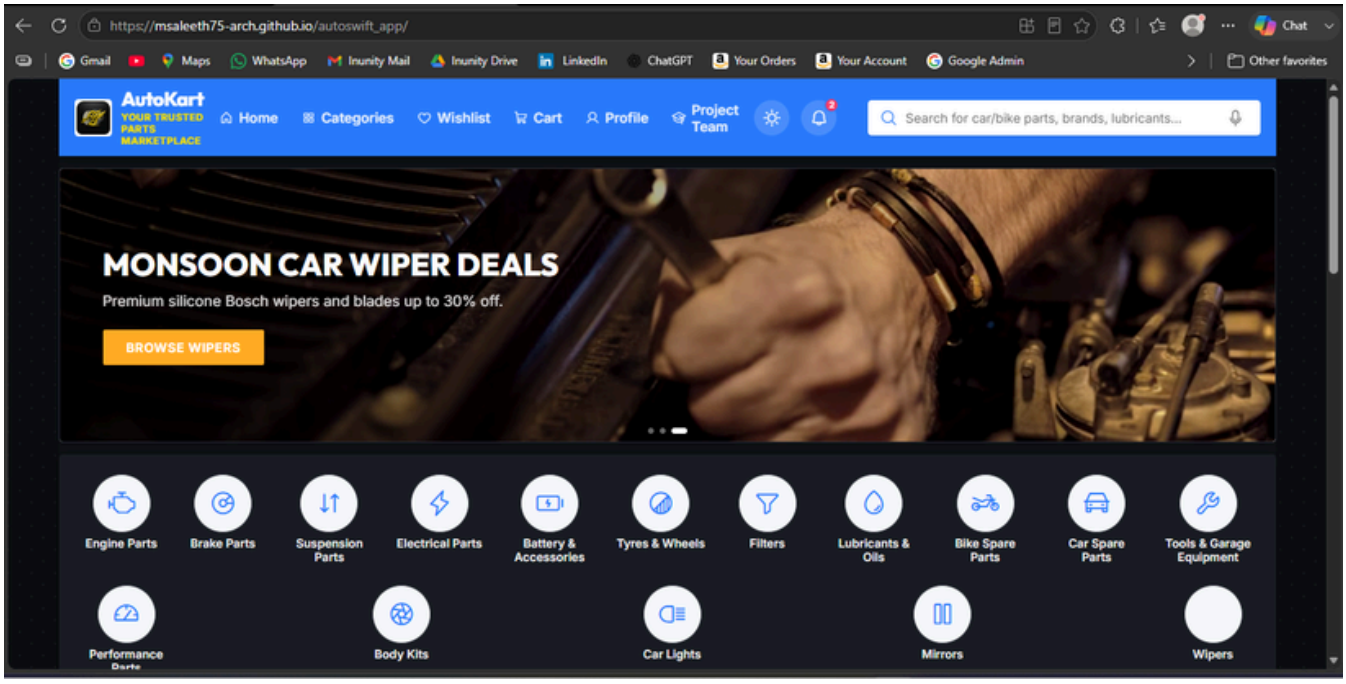
SMART SUPPLY CHAIN FOR AUTO COMPONENTS IN MANGALORE

PROBLEM STATEMENT

Vehicle owners often face difficulty in finding genuine automobile spare parts at affordable prices. Existing platforms lack a dedicated marketplace for comparing products, checking availability, and purchasing auto parts easily. This leads to delays, higher costs, and inconvenience for customers.

IDEA GENERATION AND PROTOTYPE

To solve this problem, we developed AutoKart, an online automobile spare parts marketplace. The platform allows users to browse categories, search for spare parts, compare products, view prices, add items to cart, and manage purchases efficiently.



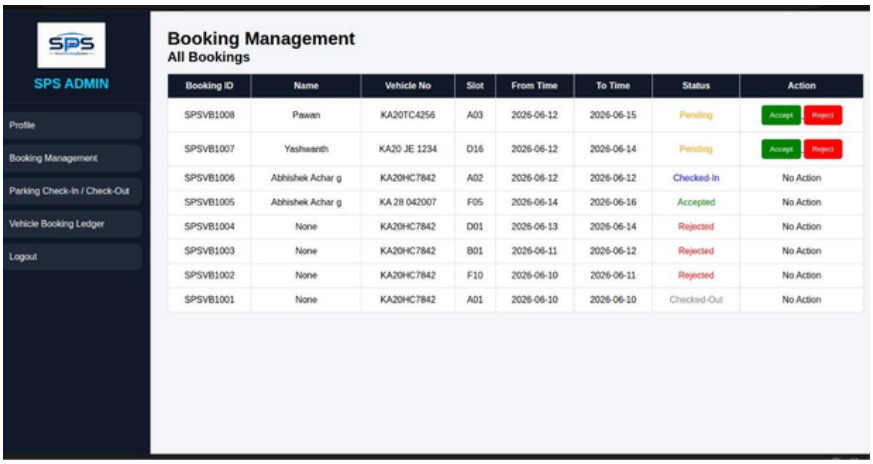
CONCLUSION

The Smart Supply Chain for Auto Components in Mangalore offers an efficient and technology-driven solution for managing automotive supply chains. By providing real-time visibility into inventory and logistics operations, the system reduces delays, minimizes stock shortages, and improves coordination among suppliers and manufacturers. The platform enhances operational efficiency, reduces costs, and supports data-driven decision-making, contributing to a more reliable and sustainable automotive supply chain ecosystem.

SMART PARKING

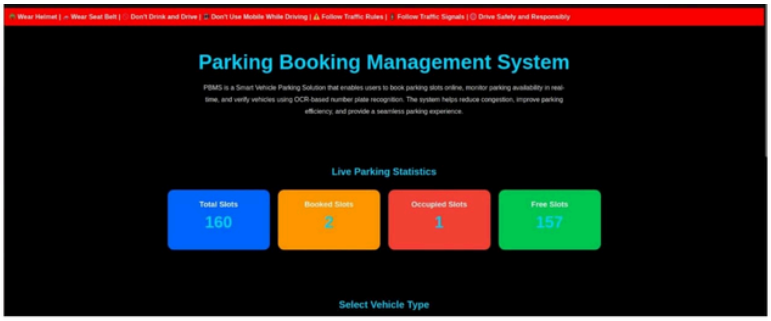
PROBLEM STATEMENT

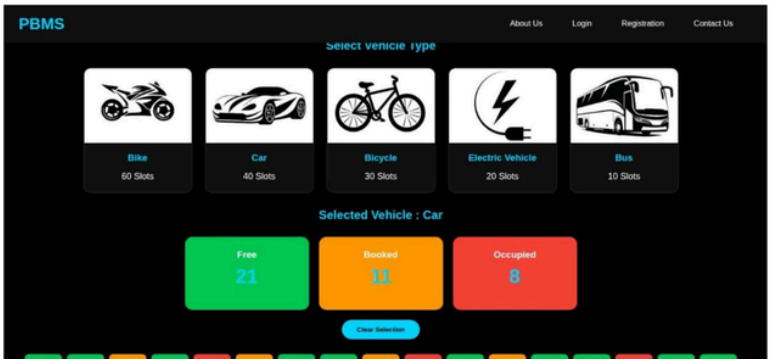
Finding a parking space in crowded urban areas is often time-consuming and frustrating. Drivers frequently spend considerable time searching for available parking slots, leading to traffic congestion, fuel wastage, and increased stress. Traditional parking systems lack real-time availability information and efficient slot management, resulting in poor utilization of parking resources.



IDEA GENERATION AND PROTOTYPE

Smart Parking is an intelligent parking management platform designed to simplify parking space discovery and reservation. The system allows users to view real-time parking availability, search for nearby parking locations, reserve slots in advance, and make digital payments through a mobile application. GPS integration helps users navigate to the selected parking area, while QR code verification enables secure entry and exit. The prototype consists of a Flutter-based mobile application, a Node.js backend, MongoDB database, GPS services, and real-time slot monitoring. Key modules include Parking Search & Booking, Slot Availability Dashboard, QR-Based Entry System, Digital Payment Gateway, User Management, and Admin Monitoring Dashboard.





CONCLUSION

The Smart Parking System provides an efficient and user-friendly solution to urban parking challenges. By offering real-time parking information, advance reservations, and digital payment facilities, the system reduces traffic congestion, saves time, and improves parking space utilization. The platform enhances convenience for drivers while enabling parking operators to manage resources more effectively, contributing to smarter and more sustainable urban transportation.

PROJECT - 33

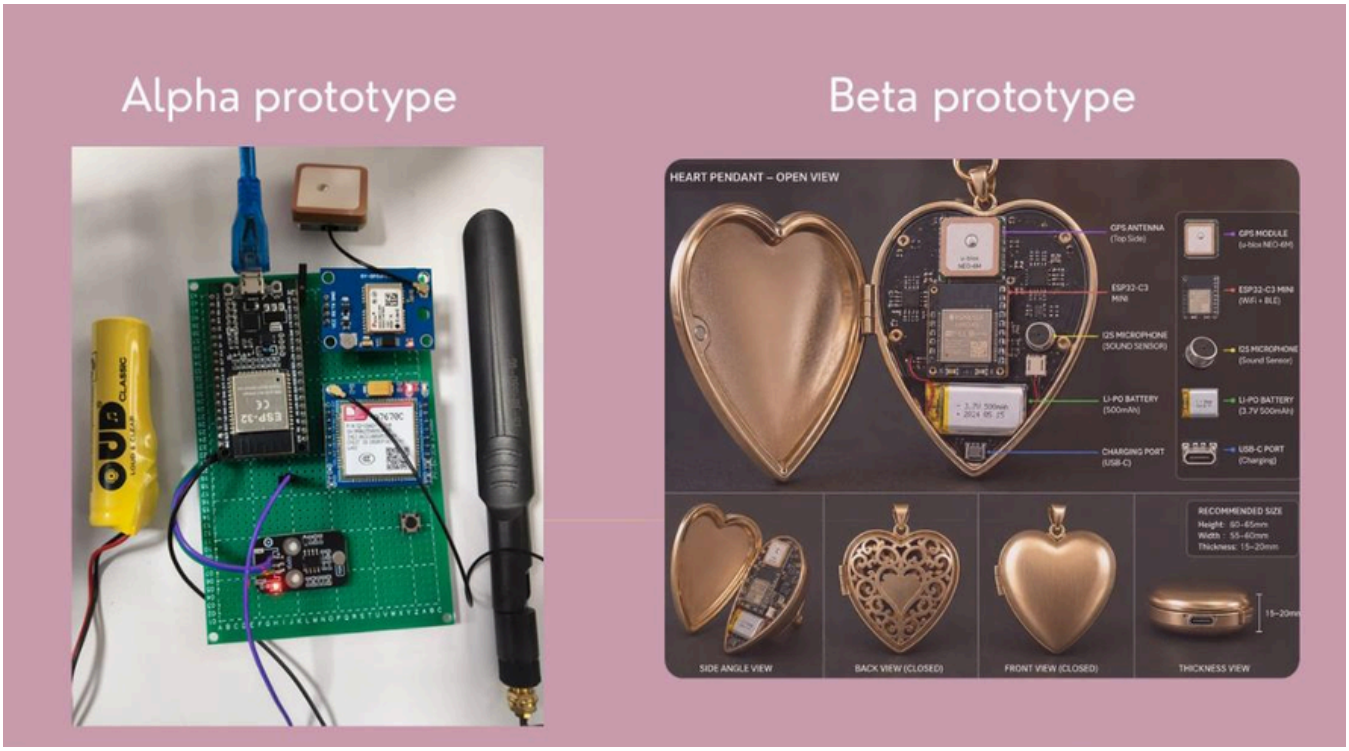
DOMESTIC VIOLENCE AND HARASSMENT OF WOMEN AT HOME RISING

PROBLEM STATEMENT

Women often face harassment, domestic violence, and other safety threats where immediate help may not be available. Existing safety solutions largely depend on smartphones and manual operation, which may not be practical during emergencies. Therefore, there is a need for a wearable safety device that can quickly detect emergencies, track location, and send alerts to trusted contacts.

IDEA GENERATION AND PROTOTYPE

The idea was developed to provide a portable and easy-to-use safety solution for women during emergency situations. After studying existing safety applications and wearable devices, a Smart Women Safety Locket was proposed. The prototype was built using an ESP32 microcontroller, GPS module, GSM module, sound sensor, SOS button, and buzzer. When a distress sound is detected or the SOS button is pressed, the system retrieves the user's location and sends emergency SMS alerts with location details to predefined contacts.



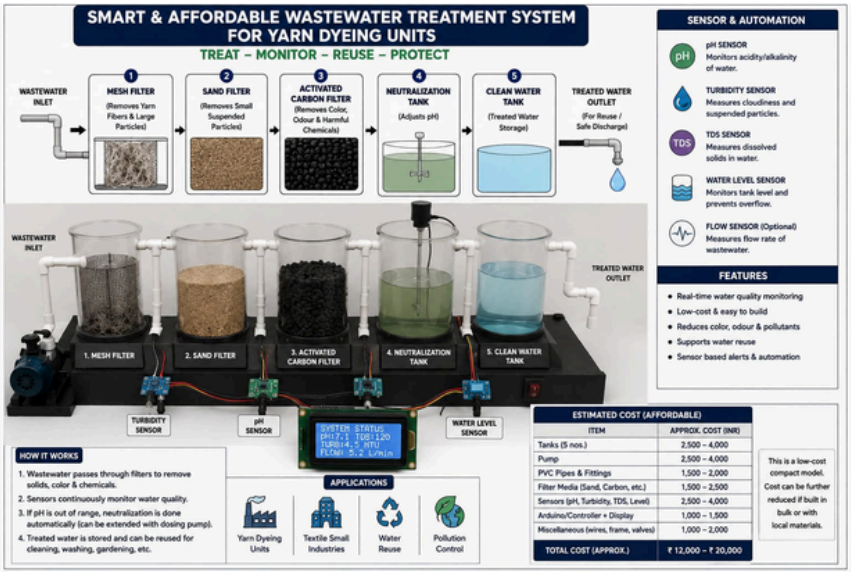
CONCLUSION

The SafeHer Smart Women Safety Locket was successfully designed and tested to provide quick emergency assistance. The system effectively detected emergency situations, obtained GPS location data, and transmitted alert messages to emergency contacts. The project demonstrates a reliable, portable, and cost-effective solution that can enhance women's safety and security during critical situations.

UNTREATED COLOURED WASTEWATER FROM YARN-DYEING UNITS

PROBLEM STATEMENT

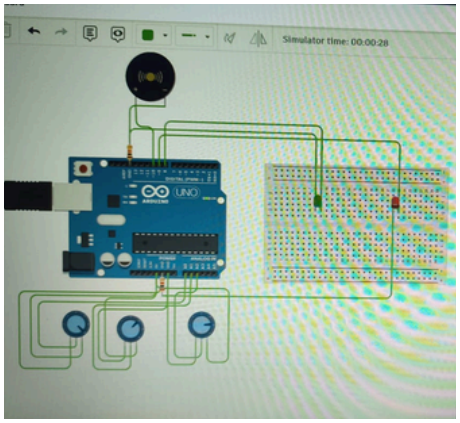
Yarn dyeing industries generate large amounts of coloured wastewater containing dyes, chemicals, and suspended impurities. Conventional wastewater treatment systems are often expensive and difficult for small-scale industries to implement. This results in environmental pollution and wastage of water resources, creating a need for an affordable and efficient wastewater treatment solution.

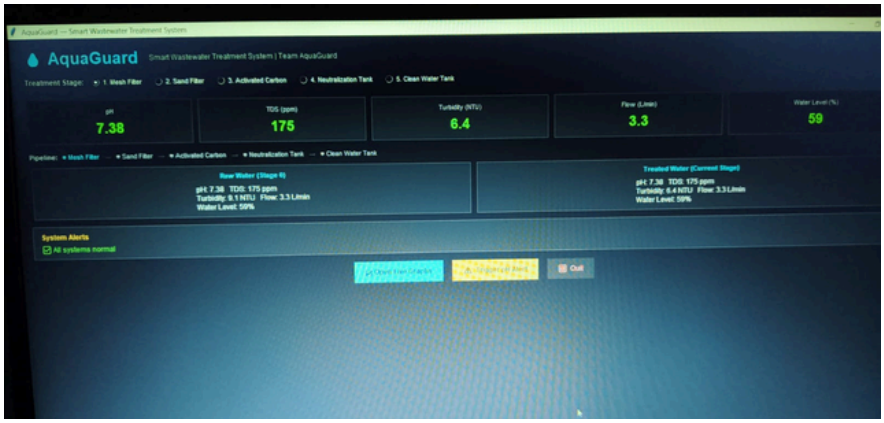


IDEA GENERATION AND PROTOTYPE

The idea was developed to create a low-cost wastewater treatment system suitable for small-scale yarn dyeing industries. Various treatment methods such as chemical treatment, biological treatment, filtration systems, and sensor-based monitoring were evaluated. After considering cost, effectiveness, and ease of maintenance, a multi-stage filtration system combined with activated carbon adsorption was selected.

The prototype consists of a collection tank, sedimentation chamber, sand and gravel filtration unit, activated carbon chamber, and treated water tank. Wastewater passes through each stage, where suspended solids, colour, odour, and impurities are gradually reduced before the treated water is collected.





CONCLUSION

The Smart Wastewater Treatment System was successfully designed and tested as a low-cost solution for treating wastewater from yarn dyeing industries. The system effectively reduced colour, turbidity, odour, and suspended impurities through sedimentation, filtration, and activated carbon adsorption. The project demonstrates a practical and environmentally friendly approach to wastewater management and water conservation for small-scale industries.

PROJECT - 35

ROOT CAUSE ANALYSIS OF AI MODEL SELECTION TRUST ISSUES

PROBLEM STATEMENT

Traditional systems store data in centralized databases, making them vulnerable to unauthorized modifications and data integrity issues. There is a need for a secure and transparent method of storing information where records cannot be altered once they are created. This project addresses this challenge by implementing a basic blockchain system that demonstrates secure and immutable data storage.

IDEA GENERATION AND PROTOTYPE

The idea was inspired by the increasing use of blockchain technology for maintaining secure and tamper-resistant records. The project aims to demonstrate the fundamental working principles of a blockchain, including block creation, hash generation, and chain validation.

The prototype allows users to enter data through a simple interface. Each data entry is stored as a block containing the information, timestamp, previous block hash, and a newly generated SHA-256 hash. Blocks are linked together to form a blockchain, ensuring that any modification to stored data can be easily detected. The system is developed using Java, HTML, CSS, and Maven to provide a simple demonstration of blockchain functionality.

CONCLUSION

The project successfully demonstrates the core concepts of blockchain technology, including block creation, cryptographic hashing, and chain integrity. By linking blocks through hash values, the system ensures that stored data remains secure and tamper-evident. The developed prototype provides a practical understanding of blockchain fundamentals and serves as a foundation for future applications involving secure and transparent data management.

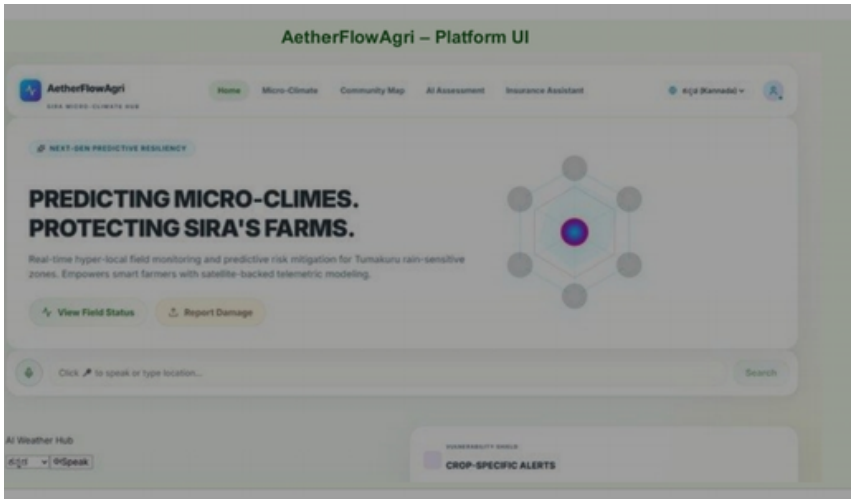
PROJECT - 36

CROP LOSS DUE TO UNSEASONAL WEATHER

PROBLEM STATEMENT

Agriculture is highly dependent on weather conditions, and unseasonal weather events such as unexpected rainfall, droughts, heatwaves, and storms can cause significant crop damage and financial losses for farmers. Most farmers lack access to timely weather alerts and data-driven insights, making it difficult to take preventive measures. This results in reduced crop yield, increased production costs, and economic instability. Therefore, there is a need for a smart system that helps farmers monitor weather conditions and receive early warnings to minimize crop losses.

IDEA GENERATION AND PROTOTYPE



The Crop Loss Prevention System is a smart agriculture platform designed to help farmers reduce losses caused by unseasonal weather conditions. The system integrates weather forecasting data, crop information, and AI-based risk analysis to predict potential threats to crops. Farmers can receive real-time alerts regarding heavy rainfall, extreme temperatures, drought conditions, or storms and obtain recommendations on protective measures. Key features include Weather Monitoring Dashboard, Crop Risk Analysis, Alert Notification System, Preventive Action Recommendations, and Farmer Advisory Services.

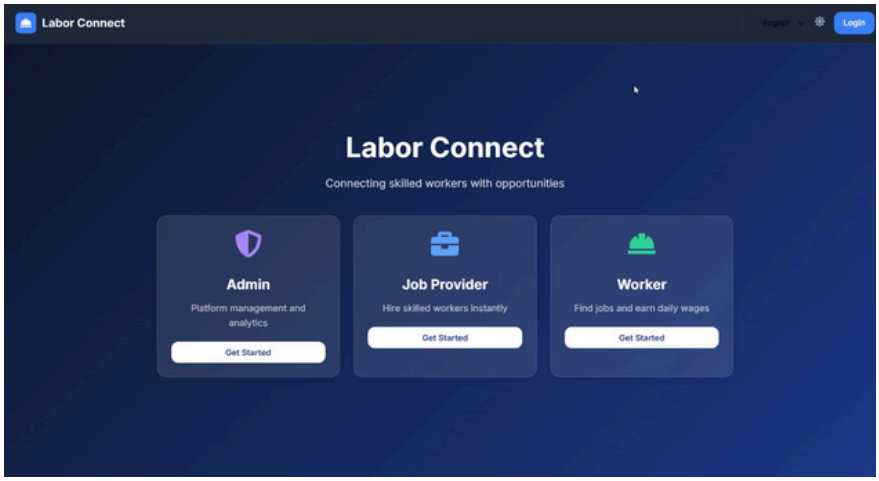
CONCLUSION

The Crop Loss Prevention System provides an effective solution for reducing agricultural losses caused by unpredictable weather conditions. By combining real-time weather monitoring, AI-based risk assessment, and early warning alerts, the system enables farmers to make informed decisions and protect their crops. The platform improves agricultural productivity, reduces financial risks, and supports sustainable farming practices, ultimately contributing to better livelihoods for farmers and enhanced food security.

EMPOWERING LABOUR AND HELPING SOCIETY

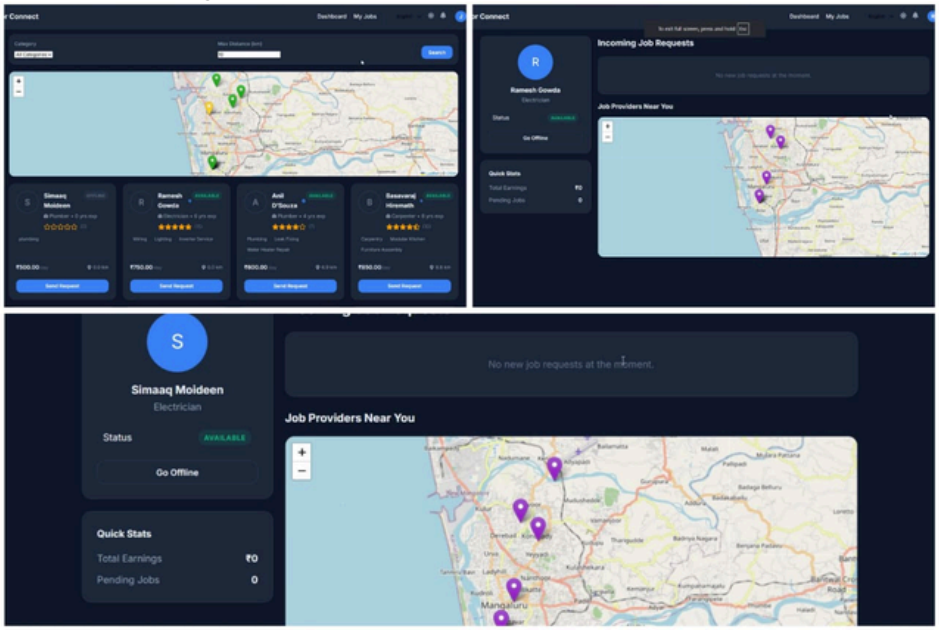
PROBLEM STATEMENT

Many daily wage laborers and unorganized workers face challenges such as limited employment opportunities, lack of access to welfare schemes, delayed payments, and poor communication with employers. At the same time, individuals and organizations seeking skilled or unskilled labor often struggle to find reliable workers quickly.



IDEA GENERATION AND PROTOTYPE

Empowering Labour and Helping Society is a digital platform designed to bridge the gap between laborers, employers, and social welfare organizations. The system enables workers to create profiles, showcase their skills, search for jobs, and receive employment notifications. Employers can post job requirements, hire workers, track work status, and process payments through the platform. Additionally, the system provides information about government welfare schemes, skill development programs, healthcare support, and emergency assistance services.

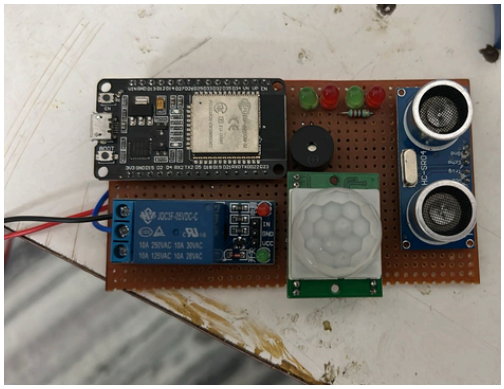


CONCLUSION

The Empowering Labour and Helping Society platform provides an effective solution for improving employment accessibility and social welfare support for laborers. By connecting workers with job opportunities, promoting skill development, and providing access to welfare resources, the system enhances economic empowerment and social inclusion. The platform benefits both laborers and employers by creating a transparent, efficient, and technology-driven ecosystem that contributes to community development and improved livelihoods.

SMART GROUND VIBRATION ANIMAL REPELLER

PROBLEM STATEMENT



Wild animals such as elephants, wild boars, deer, and monkeys frequently enter agricultural fields and damage crops, causing significant financial losses for farmers. Existing protection methods like electric fences, traps, and sound-based deterrents are often expensive, harmful to wildlife, or ineffective over time. Therefore, there is a need for a smart, eco-friendly, and automated system that can safely repel animals while protecting agricultural land.

IDEA GENERATION AND PROTOTYPE

The idea was developed to create a non-harmful animal repelling system using AI and ground vibration technology. Various animal intrusion prevention methods were studied, and an AI-assisted vibration-based approach was selected for its safety and effectiveness. The prototype was developed using an ESP32 microcontroller, PIR sensor, ultrasonic sensor, vibration motor module, relay module, and AI-based detection logic. The sensors detect animal movement, the ESP32 processes the data, and the system activates controlled ground vibrations to safely deter animals from entering the protected area.

CONCLUSION

It was successfully designed and tested as a smart and eco-friendly farm protection system. The prototype accurately detected animal movement and generated ground vibrations to repel animals without causing harm. The project demonstrates a low-cost, sustainable, and practical solution for reducing crop damage and minimizing human-wildlife conflicts.

PROJECT - 39

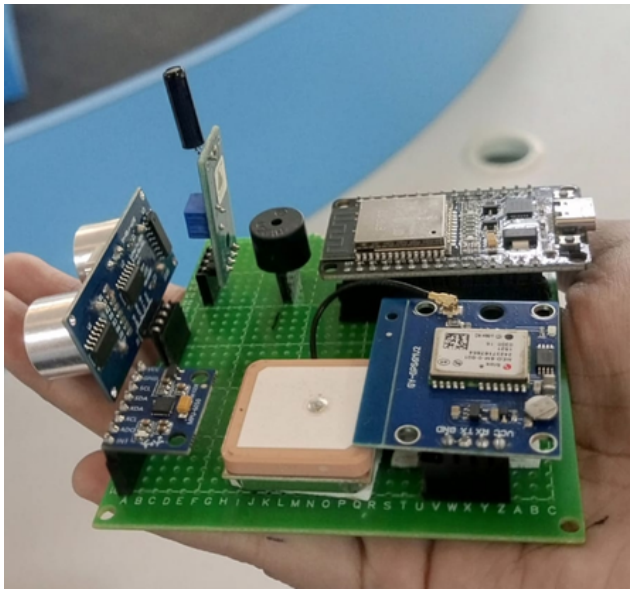
ROAD ACCIDENTS OFTEN GO UNNOTICED FOR CRITICAL MINUTES, DELAYING EMERGENCY RESPONSE AND INCREASING FATALITIES

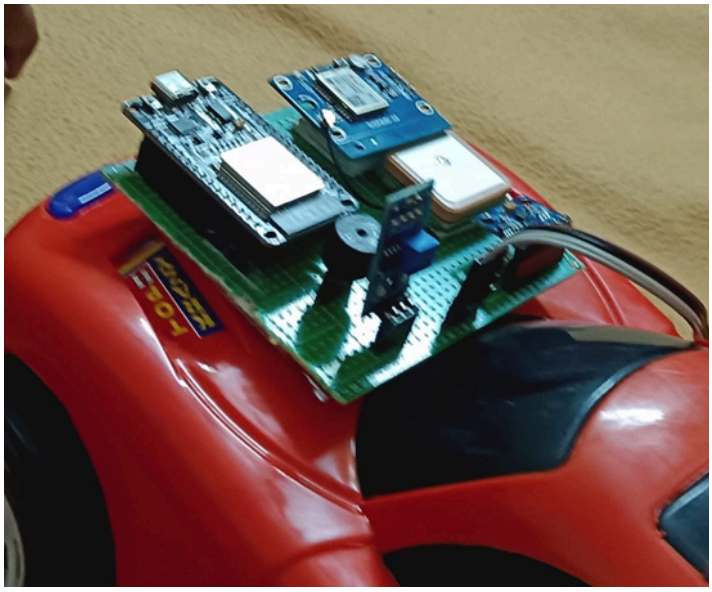
PROBLEM STATEMENT

Road accidents often result in serious injuries and loss of life because emergency services and nearby people are not informed immediately. Delays in accident detection and location reporting can slow down rescue operations and medical assistance. Therefore, there is a need for a low-cost system that can automatically detect accidents and provide real-time location information for faster emergency response.

IDEA GENERATION AND PROTOTYPE

The idea was developed to improve road safety through automatic accident detection and alert generation. Various approaches involving tilt detection, vibration sensing, obstacle detection, and location tracking were evaluated before selecting a sensor-based solution. The prototype was developed using an ESP32 microcontroller, MPU6050 sensor, vibration sensor, ultrasonic sensor, NEO-6M GPS module, and buzzer. The system continuously monitors vehicle conditions, detects accident-like events based on sensor data, retrieves the accident location, and generates emergency alerts.





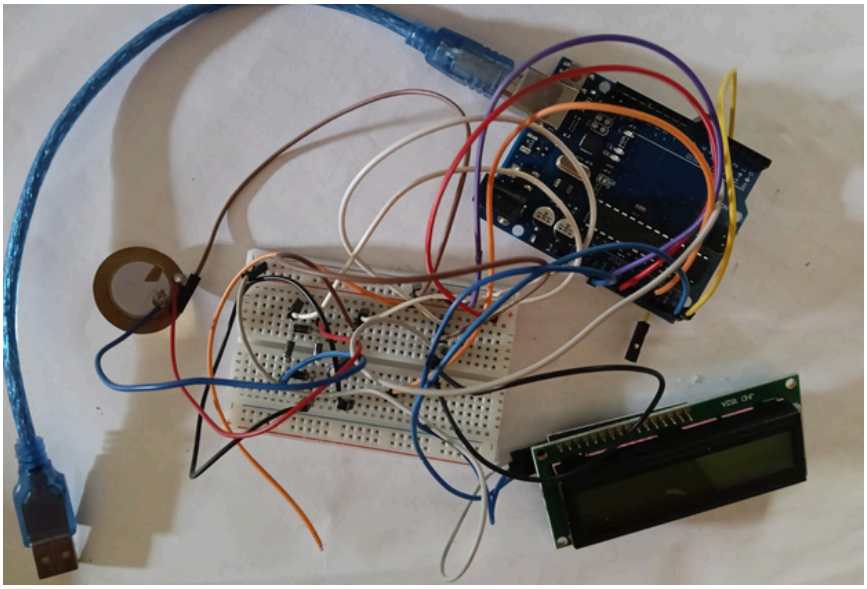
CONCLUSION

The Accident Detection and Alert System was successfully designed and tested using ESP32 and multiple sensors. The prototype accurately detected accident-like conditions, obtained location information, and generated alerts in real time. The project demonstrates a reliable, low-cost, and effective solution for improving road safety and reducing emergency response time.

SMART GREEN TUNNEL

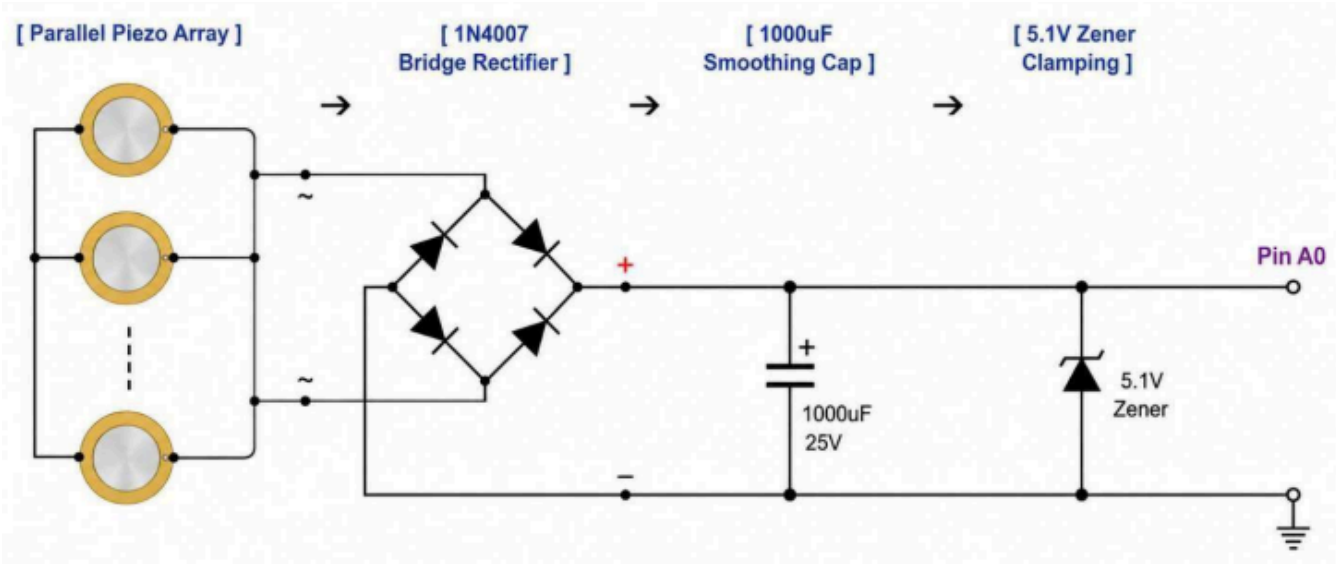
PROBLEM STATEMENT

Modern transportation tunnels consume large amounts of electricity for continuous lighting and ventilation, while the kinetic energy generated by moving vehicles is wasted as vibration, friction, and heat. Existing tunnel infrastructures do not utilize this wasted energy or adapt energy consumption based on real-time traffic conditions, leading to higher operational costs and energy inefficiency.



IDEA GENERATION AND PROTOTYPE

The idea was developed to create a self-sustaining smart tunnel system that harvests energy from vehicle movement and uses it for intelligent tunnel automation. Different harvesting methods and structural designs were evaluated, and a horizontal compression strip using piezoelectric discs was selected. The prototype was developed using piezoelectric sensors, a bridge rectifier circuit, capacitor and Zener diode conditioning network, an ATmega328P microcontroller, and an LCD display. The system converts vehicle-induced vibrations into usable electrical energy and automatically controls tunnel lighting based on vehicle detection.



CONCLUSION

The Smart Green Tunnel prototype successfully demonstrated the feasibility of harvesting kinetic energy from vehicle movement and using it for intelligent tunnel automation. The system effectively converted mechanical energy into stable electrical output, enabled real-time traffic monitoring, and dynamically controlled tunnel lighting. The project presents a sustainable and energy-efficient approach toward future smart infrastructure development.

PROJECT - 41

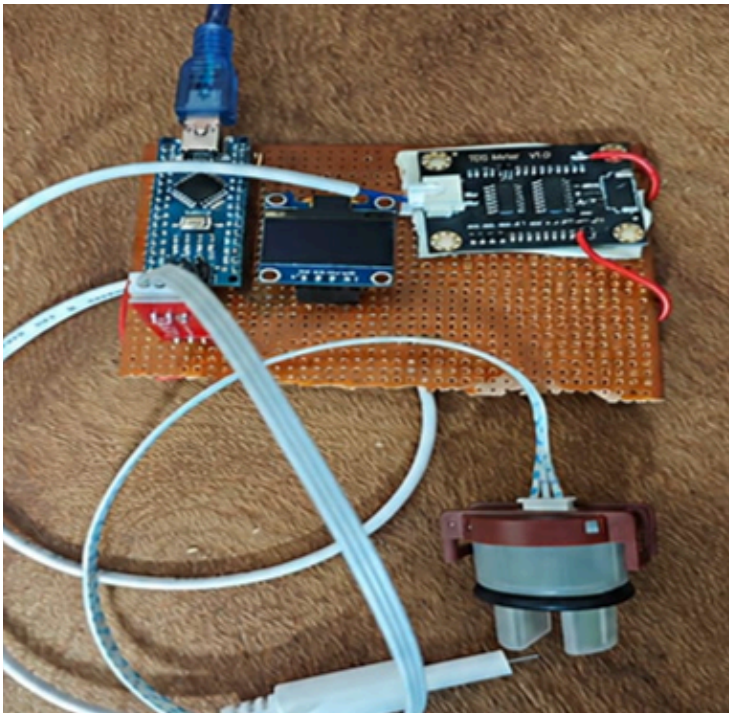
UNSAFE DRINKING WATER

PROBLEM STATEMENT

Access to safe drinking water remains a major challenge, especially in rural areas, schools, trekking locations, and disaster-affected regions. Existing water quality testing methods are often expensive, time-consuming, or require technical expertise, making it difficult for ordinary users to determine whether water is safe to drink. AquaSense aims to provide a simple, affordable, and portable solution for instant water quality assessment.

IDEA GENERATION AND PROTOTYPE

The project team explored various water testing solutions, including chemical test strips and handheld TDS meters. After evaluating affordability, portability, ease of use, and multi-parameter detection capability, they developed AquaSense—a smart bottle cap equipped with pH, TDS, and turbidity sensors. The prototype uses an Arduino Uno, LED indicators, a buzzer, and a 3D-printed enclosure to provide a quick Safe/Unsafe verdict within seconds.



CONCLUSION

AquaSense successfully demonstrated a low-cost and user-friendly smart water quality detection system. The prototype accurately identified safe and unsafe water samples, achieved a 90% user identification accuracy rate, and provided intuitive visual feedback through LED indicators. The project shows strong potential for real-world deployment to improve access to safe drinking water.

NUTRITION TRACKING

PROBLEM STATEMENT

Many individuals struggle to maintain a healthy diet due to a lack of awareness about their daily nutritional intake. Tracking calories, proteins, carbohydrates, vitamins, and other essential nutrients manually can be difficult and time-consuming. As a result, people may develop unhealthy eating habits, nutritional deficiencies, obesity, or other health-related issues. There is a need for a simple and intelligent system that helps users monitor their nutrition and make informed dietary choices.

IDEA GENERATION AND PROTOTYPE

The Nutrition Tracking System is a mobile application designed to help users monitor their daily food intake and nutritional status. Users can log meals, track calories and nutrients, set dietary goals, and receive personalized recommendations based on their eating habits. The application utilizes a Flutter-based frontend, a Node.js backend, MongoDB for data storage, and an AI module to analyze nutritional patterns and provide health insights. Key features include meal logging, nutrition analysis, goal tracking, and progress monitoring through an interactive dashboard.

The prototype consists of a Flutter-based mobile application for a smooth user experience, a Node.js backend for processing and managing user data, MongoDB for secure data storage, and Firebase Cloud Messaging for notifications and reminders. The system also includes graphical dashboards that display daily, weekly, and monthly nutrition trends. Users can receive alerts for exceeding calorie limits, missing nutritional targets, or maintaining unhealthy eating habits. Key application pages include the Food Log Page, Nutrition Analysis Dashboard, Goal Tracking Page, Meal Recommendation Page, and User Profile Page, providing a comprehensive platform for effective nutrition management.

CONCLUSION

The Nutrition Tracking System provides an effective solution for maintaining healthy eating habits and improving overall wellness. By enabling users to monitor their nutritional intake, track dietary goals, and receive personalized recommendations, the system encourages healthier lifestyle choices. It helps prevent nutritional imbalances, supports fitness and health objectives, and promotes long-term well-being through data-driven insights and continuous monitoring.

SMART PATIENT MONITORING AND ALERT SYSTEM

PROBLEM STATEMENT

Many elderly and chronically ill patients face difficulties in managing their health at home. Common challenges include forgetting to take medications on time, limited access to healthcare facilities, lack of continuous health monitoring, poor communication among family members, and delayed medical responses during emergencies. These issues can lead to worsening health conditions and reduced quality of care. Therefore, there is a need for an intelligent healthcare monitoring system that ensures timely medication reminders, continuous patient monitoring, and real-time communication with caregivers and family members.

IDEA GENERATION AND PROTOTYPE

The proposed Smart Patient Monitoring & Alert System is designed to improve patient care through timely medication reminders, health monitoring, and family connectivity. The system allows patients to upload medical reports such as blood pressure and blood sugar readings, while an AI-based health analysis module classifies the patient's condition as Normal, Warning, or Critical and recommends whether medical attention is required. The application connects up to four family members, enabling real-time monitoring and shared access to health updates. Automated alerts are sent whenever a medication dose is missed or abnormal health data is detected. The prototype consists of a Flutter-based mobile application, a Node.js backend with real-time synchronization, MongoDB for data storage, and Firebase Cloud Messaging for notifications. Key application pages include the Medical Report Page, Dosage Tracking Page, and Patient Health Status Page.

CONCLUSION

The Smart Patient Monitoring & Alert System offers an effective solution to the healthcare challenges faced by elderly and chronically ill patients. It promotes timely medication adherence, enables remote health monitoring, reduces the need for frequent hospital visits, and improves communication among family members. By utilizing AI-based health analysis and real-time alerts, the system helps identify potential health risks early and supports timely medical intervention. Overall, the solution enhances patient safety, healthcare accessibility, and quality of life, demonstrating how technology can be leveraged to create smarter and more efficient healthcare management.

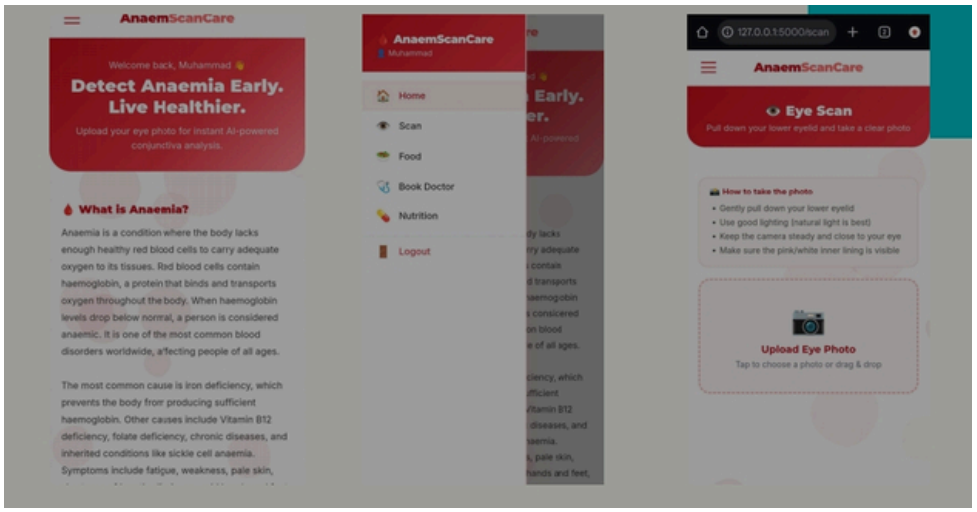
PROJECT - 44

SMART NUTRITION AND ANAEMIA MONITORING SYSTEM FOR CHILDREN IN LOW INCOME FAMILIES

PROBLEM STATEMENT

Anaemia and poor nutrition are major health concerns among children from low-income families. Limited access to nutritious food, lack of regular health monitoring, and low awareness about dietary requirements often lead to iron deficiency, malnutrition, and delayed physical and cognitive development. Early detection of anaemia is frequently overlooked due to inadequate healthcare access and irregular health checkups. Therefore, there is a need for an affordable and intelligent system that helps monitor nutrition levels and identify anaemia risks at an early stage.

IDEA GENERATION AND PROTOTYPE



The Smart Nutrition and Anaemia Monitoring System is a digital healthcare platform designed to track children's nutritional status and assess the risk of anaemia. The system allows caregivers or health workers to record health parameters such as age, weight, height, hemoglobin levels, and dietary intake. An AI-based analysis engine evaluates the collected data to identify nutritional deficiencies and classify anaemia risk levels as Normal, Moderate, or High Risk.

CONCLUSION

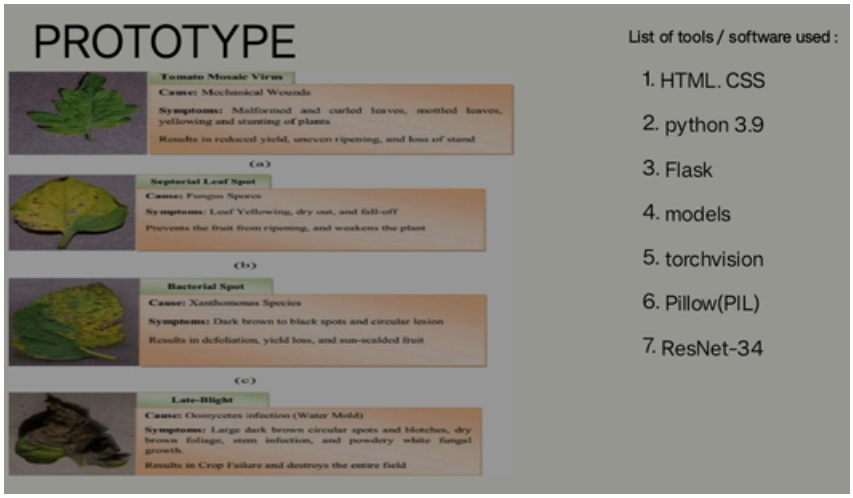
The Smart Nutrition and Anaemia Monitoring System provides an effective solution for improving child health in low-income communities. By enabling continuous nutrition tracking, early anaemia detection, and personalized dietary recommendations, the system supports better health outcomes and preventive care. The platform enhances awareness among caregivers, facilitates timely interventions, and contributes to reducing malnutrition and anaemia among children through technology-driven healthcare monitoring.

LEAF CLASSIFICATION FOR ITS MEDICINAL USE AND DISEASE DETECTION

PROBLEM STATEMENT

Medicinal plants play an important role in traditional and modern healthcare, but identifying plant species and detecting leaf diseases often requires expert knowledge. Farmers, students, and the general public may find it difficult to distinguish between medicinal plants or recognize early signs of plant diseases. Incorrect identification can lead to improper usage of medicinal plants and delayed treatment of plant infections, resulting in reduced plant health and productivity.

IDEA GENERATION AND PROTOTYPE



The Leaf Classification for Medicinal Use and Disease Detection System is an AI-powered application that identifies plant species from leaf images and determines their medicinal benefits while simultaneously detecting possible leaf diseases. Users can upload or capture leaf images using a mobile device, and the system uses deep learning and image processing techniques to classify the plant and analyze its health condition. The system provides plant information, medicinal uses, disease diagnosis, treatment recommendations, and a history of analyzed leaves. Key modules include Leaf Image Upload, Plant Classification Engine, Disease Detection Module, Medicinal Information Dashboard, and Recommendation System.

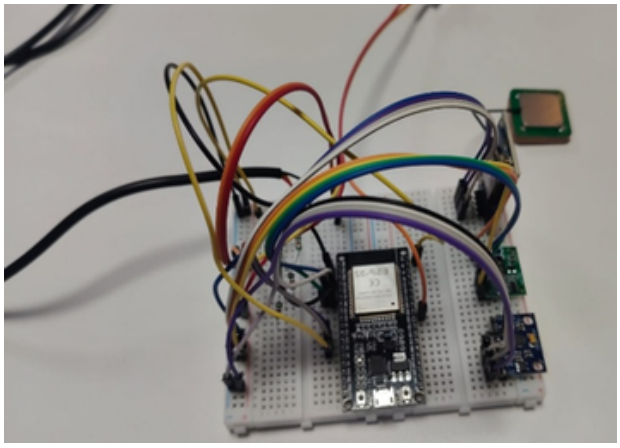
CONCLUSION

The Leaf Classification for Medicinal Use and Disease Detection System offers an efficient and accessible solution for plant identification and health monitoring. By combining AI-based image recognition with medicinal plant knowledge, the system helps users accurately identify plants, understand their medicinal value, and detect diseases at an early stage. This promotes better plant care, supports agricultural productivity, and increases awareness of the medicinal benefits of various plant species through technology-driven insights.

LIVE STOCK HEALTH MANAGEMENT SYSTEM

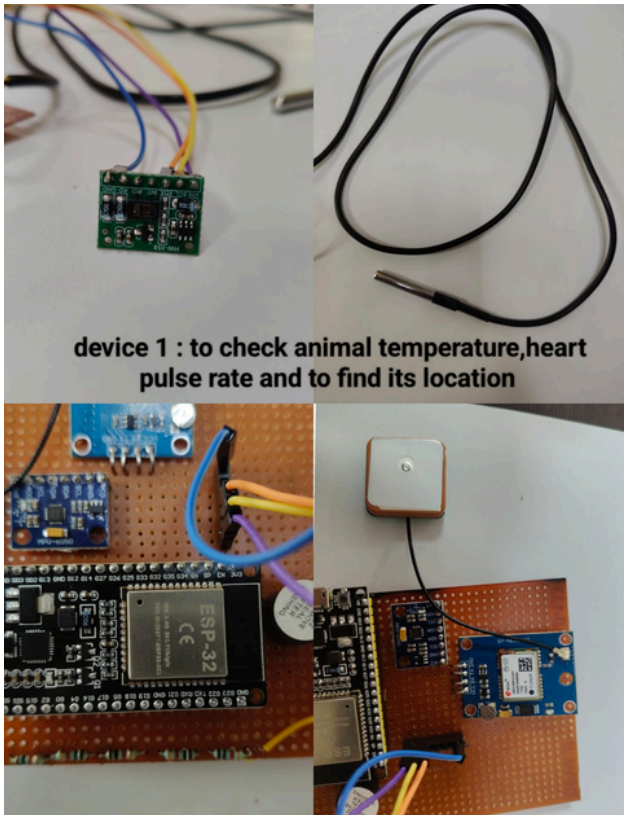
PROBLEM STATEMENT

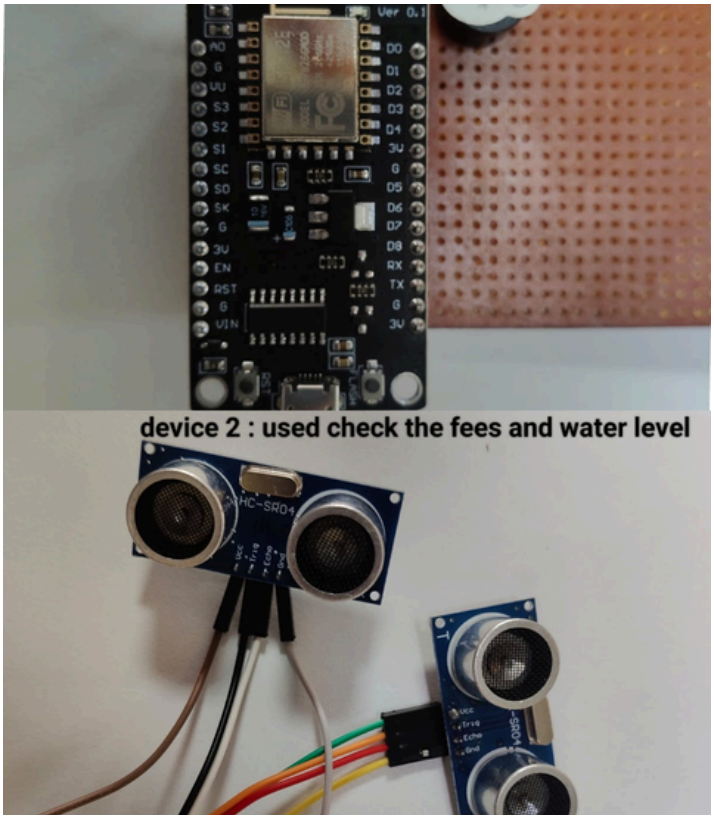
Farmers face difficulties in continuously monitoring the health, location, feed, and water availability of their livestock. Manual monitoring is time-consuming and can lead to delayed detection of health issues, animal loss, and inadequate feed or water supply. Therefore, there is a need for a smart and cost-effective system that can provide real-time monitoring and alerts for better livestock management.



IDEA GENERATION AND PROTOTYPE

The idea was developed to help farmers monitor livestock health and safety using IoT technology. The prototype was built using ESP32, NodeMCU, DS18B20 temperature sensor, MAX30102 pulse sensor, NEO-6M GPS module, ultrasonic sensors, and Firebase. The system monitors body temperature, heart rate, animal location, feed levels, and water levels in real time, while displaying data on a mobile application and generating alerts when resources become low.





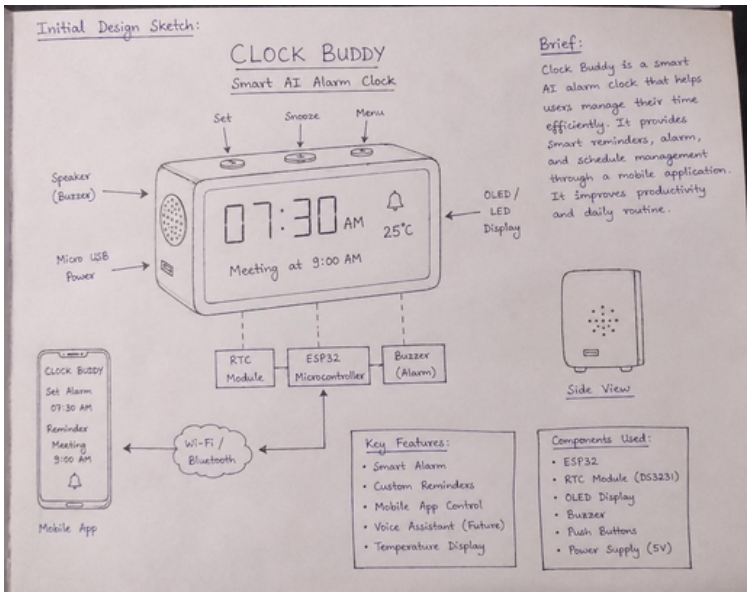
CONCLUSION

The Livestock Health Management System successfully demonstrated real-time monitoring of animal health, location, feed, and water levels. The system provided reliable data, generated timely alerts, reduced manual monitoring efforts, and improved livestock management efficiency. The project demonstrates a practical and affordable IoT-based solution for enhancing animal welfare and farm productivity.

SMART AI ALARM CLOCK

PROBLEM STATEMENT

Many students and working professionals face difficulties in waking up on time, remembering important tasks, and maintaining a consistent daily routine. Traditional alarm clocks provide only basic alarm functionality and do not offer intelligent reminders or scheduling assistance. As a result, users may experience poor time management, reduced productivity, and increased stress. Therefore, there is a need for a Smart AI Alarm Clock that combines alarms, reminders, and scheduling support in a single user-friendly system.



IDEA GENERATION AND PROTOTYPE

The idea was developed to help users improve time management, productivity, and daily routine planning. After evaluating various solutions such as mobile reminder applications, voice assistants, and smart schedulers, a Smart AI Alarm Clock was selected as the most practical and cost-effective option. The prototype was developed using an ESP32 microcontroller, DS3231 RTC module, OLED display, buzzer, push buttons, and AI-based reminder logic. The system provides accurate alarm notifications, task reminders, and basic scheduling assistance through an interactive display and alert mechanism.

CONCLUSION

The prototype successfully improved alarm and reminder functionality, offering a reliable and user-friendly solution for better time management.

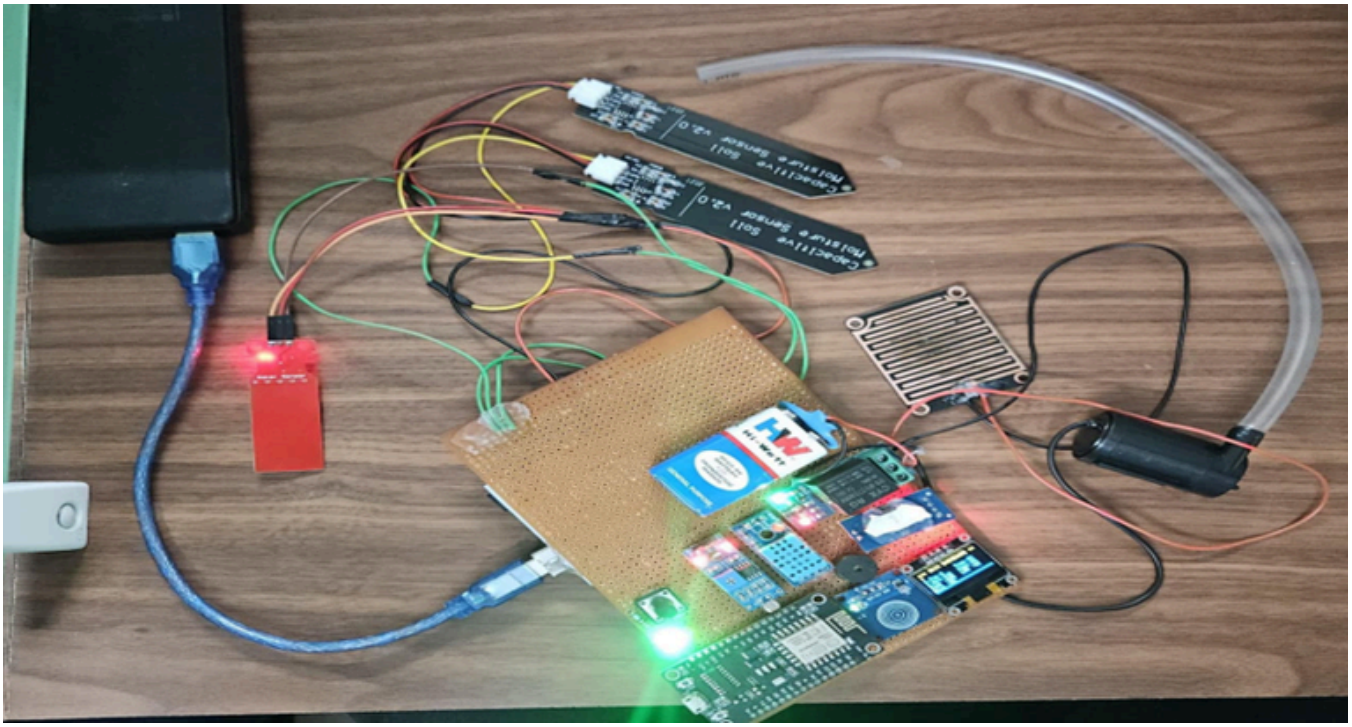
POOR CROP PRODUCTION AND WATER WASTAGE

PROBLEM STATEMENT

Agriculture heavily depends on proper irrigation, but traditional irrigation systems require continuous human supervision and often result in overwatering or underwatering of crops. This leads to water wastage, increased labor, and reduced agricultural productivity. Therefore, there is a need for a low-cost automatic irrigation system that monitors soil conditions and supplies water only when required.

IDEA GENERATION AND PROTOTYPE

Several irrigation approaches, including manual irrigation, timer-based irrigation, and sensor-based irrigation, were evaluated. Sensor-based irrigation was selected due to its higher efficiency and water conservation capabilities. The prototype was developed using an Arduino Uno, dual soil moisture sensors, DHT11 temperature and humidity sensor, relay module, water pump, buzzer, and OLED display. The system continuously monitors soil moisture and automatically controls the water pump based on predefined moisture thresholds.



CONCLUSION

The Dual Sensor Smart Irrigation System was successfully designed and implemented as an automated irrigation solution. The system accurately detected soil moisture levels, controlled the water pump efficiently, reduced water wastage, and minimized manual intervention. The project demonstrated a practical, economical, and environmentally friendly approach to smart farming and can be further enhanced using IoT, solar power, and AI-based irrigation prediction technologies.

PROJECT - 49

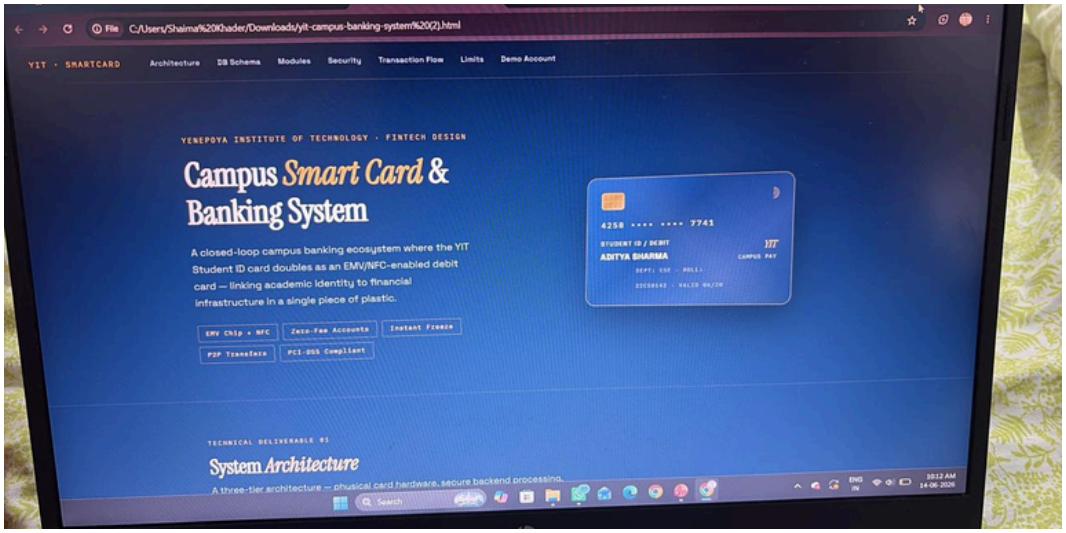
CAMPUS SMART CARD AND BANKING SYSTEM

PROBLEM STATEMENT

Many educational institutions still rely on separate systems for student identification, attendance tracking, fee payments, library access, and campus services. This often leads to inefficiencies, manual record management, security concerns, and inconvenience for students and administrators. Students frequently need multiple cards or credentials to access different services, while institutions face challenges in maintaining centralized records. Therefore, there is a need for a unified digital solution that integrates campus services with secure banking and payment functionalities.

IDEA GENERATION AND PROTOTYPE

The Campus Smart Card and Banking System is an integrated digital platform that combines student identification, campus access, and financial transactions into a single smart card solution. The system enables students to use a smart card or mobile application for attendance marking, library services, hostel access, cafeteria payments, fee payments, and other campus-related transactions. It also integrates with banking services to support cashless payments, balance management, and transaction history tracking.



CONCLUSION

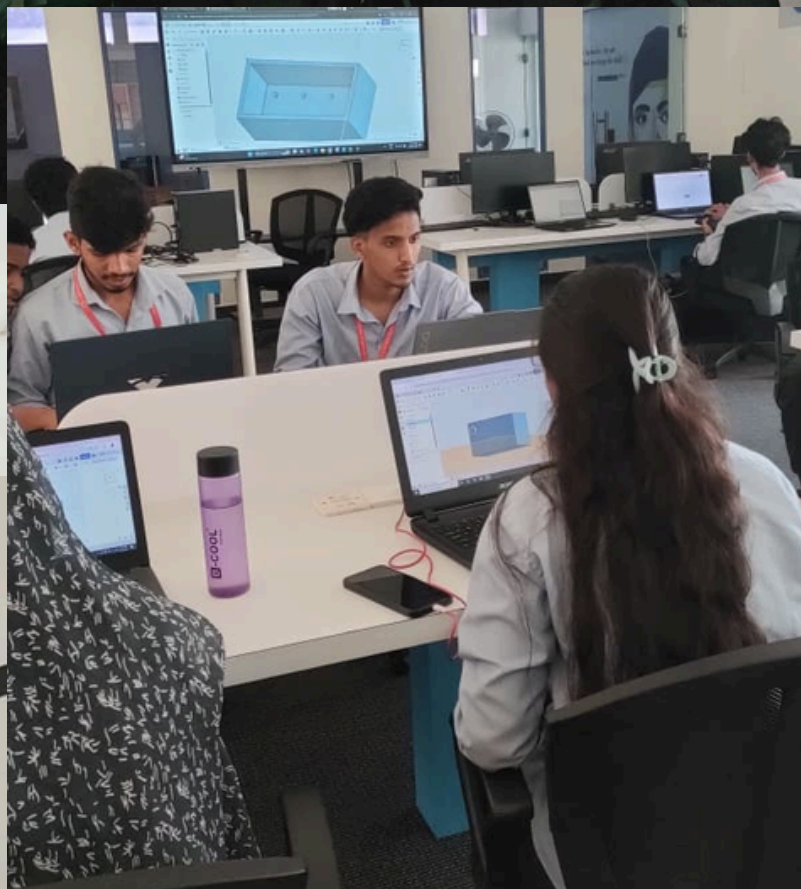
The Campus Smart Card and Banking System provides a secure, efficient, and centralized solution for managing campus services and financial transactions. By integrating identification, access control, attendance, and digital payments into a single platform, the system enhances convenience for students and improves administrative efficiency. The solution promotes a cashless campus environment, strengthens security, and simplifies the management of institutional resources through digital transformation.

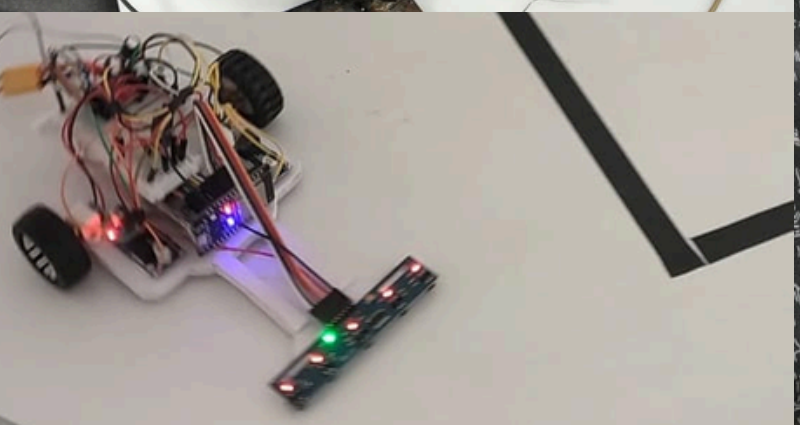
inunity
innovation for community

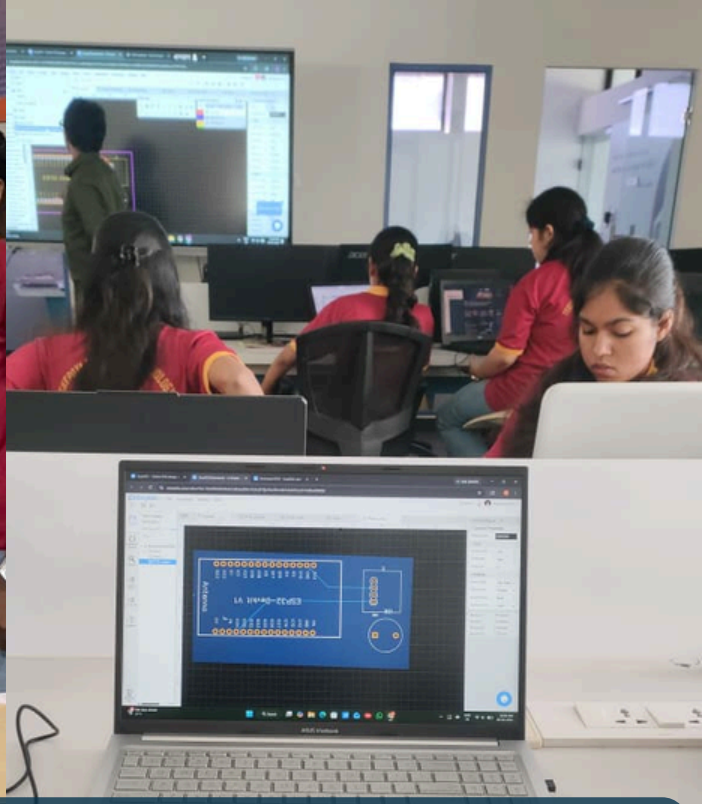
YENEPOYA

Read an Input

YENEPOYA


O-COOL





“True education begins when learning meets doing.”